

Group-level analysis

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Group-level analysis

- Estimation – in parallel across all voxels “Mass-univariate”
- Inference – uses spatial information and correlation among measurements

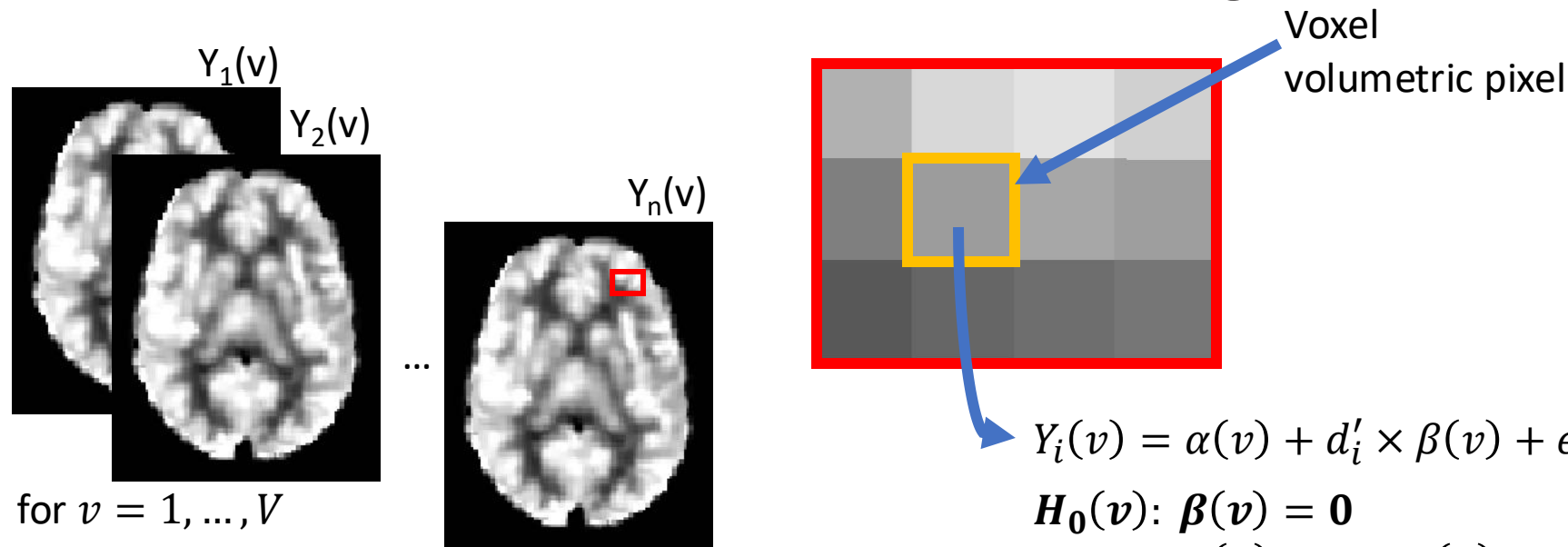
Group-level estimation

Two important features to consider in estimation:

- **Heteroskedasticity** – variance differs between participants
- **Correlation** – measurements from the same individual are correlated
- Accounting for these improves estimation efficiency
 - Also needed to obtain unbiased effect size estimates and test statistics

Mass-univariate approach

- Mass-univariate estimation
 - Widely used to localize regions of association ($\approx 18,800$ studies)
 - Perform inference at each location in the image



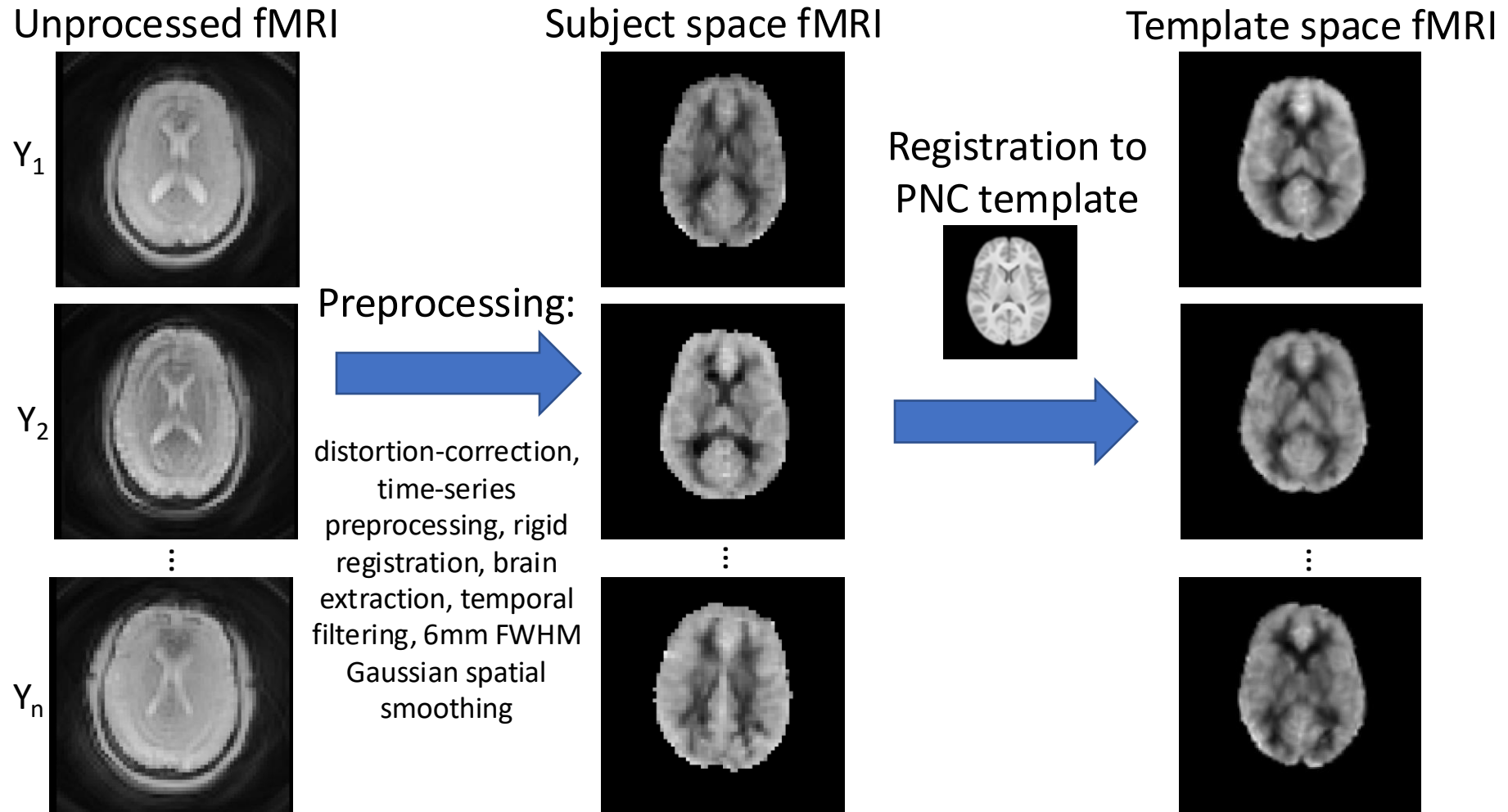
$$Y_i(v) = \alpha(v) + d'_i \times \beta(v) + \epsilon_i(v),$$

$$H_0(v): \beta(v) = 0$$

$$\text{Reject } H_0(v) \text{ where } p(v) < \alpha$$

Friston et al., *Neuroimage*, 1994
Nichols,
http://blogs.warwick.ac.uk/nichols/entry/bibliometrics_of_cluster/, 2016

fMRI preprocessing steps



Despicable Me first-level analysis



Subject-level time series model:

$$\begin{aligned}
 & \text{Time series data} = \text{Positive waveform} \times \beta_{i, \text{positive}}(v) + \text{Negative waveform} \times \beta_{i, \text{negative}}(v) \\
 & + \epsilon_i(v, t)
 \end{aligned}$$

Group-level analysis:

$$Y_i(v) := \hat{\beta}_{i, \text{positive}}(v) - \hat{\beta}_{i, \text{negative}}(v)$$

Outcome is increased activation in 2-back condition

Heteroskedasticity in group-level analysis

- *Despicable Me* data

- $Y_i(v) := \hat{\beta}_{i,positive}(v) - \hat{\beta}_{i,negative}(v)$
- It's possible that $\text{Var}\{Y_i(v) | X_i\} = g(X_i)$, which implies unequal variances, also called **heteroskedasticity/nonexchangeability**
 - E.g. motion causes increased noise
- Even worse: $\text{Cov}\{Y_i(v), Y_i(w) | X_i\} = g(X_i)$ implies nonexchangeability

- Other examples:

- Cortical thickness, functional connectivity
- Any subject-level estimates

Multilevel models in neuroimaging

- A group-level analysis might use a multivariate model

$$Y_i(v) := [\hat{\beta}_{i,positive}(v), \hat{\beta}_{i,negative}(v)]$$

- Multiple measurements per participant – requires accounting for **correlation within participant**
- There could be many conditions included in the analysis
 - Some tasks have multiple runs (e.g. they scan over three 8 min sessions). These are included as repeated measurements
- Random effects analysis

Potential data structures

- And appropriate methods

Number of Measurements	Variance Type	
	Homosked.	Heterosked.
	One	Multiple
Number of Measurements	One	Classical statistical methods
	Multiple	Variance weighted GLM; Robust standard errors (sPBJ)
Number of Measurements	One	Random effects models
	Multiple	FLAME ; GEE (sPBJ , SwE)

Options for group-level analysis

- Classical approach – using linear model at group-level
- Random effects analysis – with multiple measurements on the same participants
- FLAME – random effects **with** heteroskedasticity
- Semiparametric bootstrap joint inference (sPBJ) – research from my group. Correlated data **with** robust standard errors
 - Also implemented in SwE toolbox in SPM
- Permutation testing

Classic approach – ignores first-level model

- Assume

$$Y_i = X_i\beta + \epsilon_i$$

- $\epsilon_i \sim N(0, \sigma^2)$
- If σ_i^2 does not depend on X_i , this model is still valid – why?
- Normality assumption is not strictly necessary – why?
- What is the estimator for β and what is the variance of $\hat{\beta}$?

Robust (heteroskedasticity-consistent) standard errors approach

- Assume

$$Y_i = X_i\beta + \epsilon_i$$

- $\epsilon_i \sim N(0, \sigma_i^2)$, participants have different variances
- If σ_i^2 depends on X_i , possibly in an unknown way
- Normality assumption is not strictly necessary – why?
- What is an estimator for β and what is the variance of the estimator?
 - The least squares estimator is ok

Multi-level (random effects) model

- A random effects model is used to account for correlation among repeated measurements

- First-level time-series model, matrix equation:

$$Y_k = X_k \beta_k + \epsilon_k$$

- $\text{Cov}(\epsilon_k) = V$

- Group-level (second-level) model:

$$\beta_k^T = X_{Gk} \beta_G + \eta_k$$

- β_k - can think of this as a column vector or a scalar parameter
 - X_{Gk} is a matrix where each row is the same set of participant-level covariates (e.g. participant k 's age, sex, and task performance.
 - $\text{Cov}([\eta_1, \dots, \eta_n]) = V_G$ - for n independent subjects, η_k are assumed IID

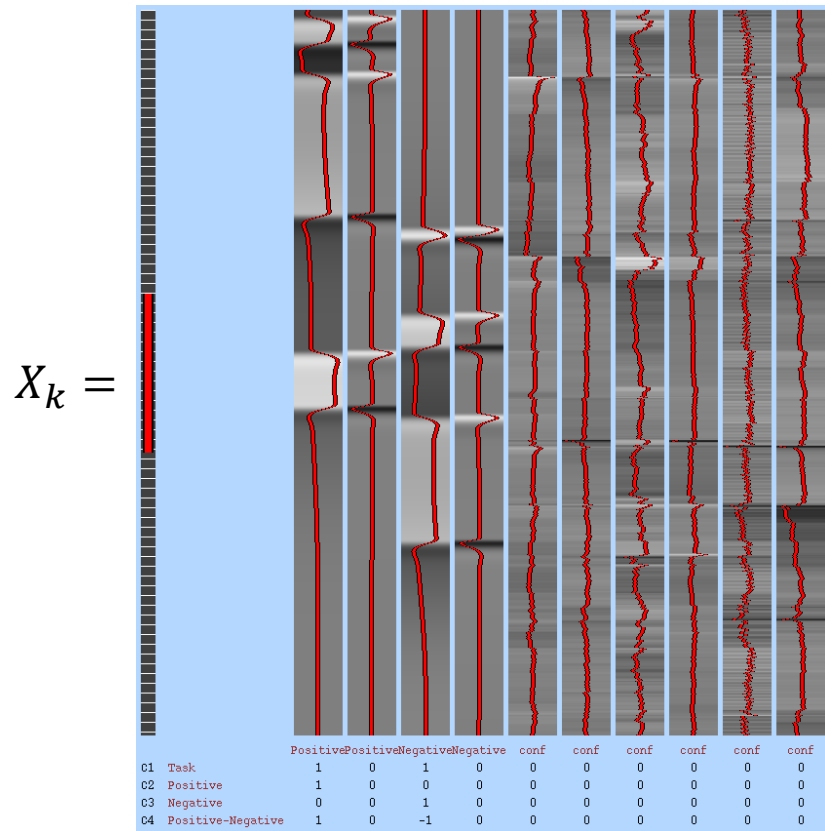
Combining the two equations

- Combining the two equations gives a typical formula for a random effects model

$$Y_k = X_k X_{Gk} \beta_G + X_k \eta_k + \epsilon_k$$

- $\epsilon_k \sim N(0, V)$ – these are the error variances (time series dependence)
- $\eta_k \sim N(0, V_\eta)$ – these are the random effect term with variance
- ϵ_k and η_k are independent

Examples of design matrices in my notation



Dims: T x 10

$$\beta_k = \begin{bmatrix} \beta_{1k} \\ \beta_{2k} \\ \vdots \\ \beta_{10,k} \end{bmatrix}$$

Dims: 1 x 10

$$X_{Gk} = \begin{bmatrix} \text{age}_k & \cdots & \text{sex}_k \\ \vdots & \ddots & \vdots \\ \text{age}_k & \cdots & \text{sex}_k \end{bmatrix}$$

Dims: 10 x p

$$\beta_G = [\beta_{G1} \quad \cdots \quad \beta_{G,10}]$$

Dims: p x 10

Notation

- I found the notation in the FLAME paper a little confusing
- My notation differs by using subject level notation and treating β_k as a vector
- In their paper they seem to collapse β_k as a single vector across all subjects (instead of a matrix), and so X_{Gk} would need to have a Kronecker product kind of shape

FLAME

- FLAME is FSL's approach that uses Bayesian estimation
 - It is a multilevel random effects model with unequal variances
 - First-level time-series model, matrix equation:
$$Y_k = X_k \beta_k + \epsilon_k$$
 - $\text{Cov}(\epsilon_k) = V_k$. Covariance being indexed by k implies what?
 - Group-level (second-level) model:
$$\beta_k^T = X_G \beta_G + \eta_k$$
 - β_k - can think of this as a column vector or a scalar parameter
 - $\text{Cov}([\eta_1, \dots, \eta_n]) = V_G$ - for n independent subjects, η_k are assumed IID.
- Notation from the reading: Beckmann paper ("flame.pdf")
- What is the difference from the random effects model?

FLAME estimation

- Estimation is complex –occurs in multiple levels, which is convenient (Section 2.C of Beckmann paper)
- More complex due to estimation of the variance terms

Semiparametric bootstrap joint (sPBJ) inference

- This is what I called the “robust approach” on a previous slide and is essentially an imaging version of GEE
- Can accommodate heteroskedasticity and longitudinal models
- Uses “sandwich” covariance estimates for asymptotically unbiased spatial covariance function estimator
- Guillaume et al., 2014 and Vandekar et al., 2019 (see homework/reading)

Estimating equations approach

- Target parameters can be written as solutions to estimating equations

$$\hat{\beta}(v) = \operatorname{argmax}_{\beta(v)} - \sum_i^n W_i(v) \left(Y_i(v) - X_i^T \beta(v) \right)^2 = \Psi(Y(v), \beta(v))$$

- Differentiating:

$$\frac{\partial \Psi}{\partial \beta(v)} = \sum_i^n W_i(v) \{ Y_i(v) - X_i^T \beta(v) \} X_i$$
$$\frac{\partial^2 \Psi}{\partial \beta^2(v)} = - \sum_i^n W_i(v) X_i X_i^T$$

Taylor expansion of estimating equation

- $\frac{\partial \Psi}{\partial \beta(v)} \{\hat{\beta}(v)\} = 0$

- Taylor expansion:

$$0 = \sum_i^n W_i(v) \{Y_i(v) - X_i^T \beta(v)\} X_i^T - \sum_i^n W_i(v) X_i X_i^T \{\hat{\beta}(v) - \beta(v)\}$$

- Which implies

$$\sqrt{n} \{\hat{\beta}(v) - \beta(v)\} = \left\{ n^{-1} \sum_i^n W_i(v) X_i X_i^T \right\}^{-1} n^{-1/2} \sum_i^n W_i(v) \{Y_i(v) - X_i^T \beta(v)\} X_i^T$$

Asymptotic joint distribution

- $n^{-1} \sum_i^n W_i(v) X_i X_i^T \rightarrow_P A(v)$
- $n^{-1/2} \sum_i^n W_i(v) \{Y_i(v) - X_i^T \beta(v)\} X_i \rightarrow_D$
 $N(0, B(v, v)),$

where $B(v, v) = \mathbb{E}\{W_i(v)^2 \{Y_i(v) - X_i^T \beta(v)\}^2 X_i^T X_i\}$

- For imaging we are particularly interested in the covariance terms
$$\text{Cov} \left\{ \sqrt{n} \left(\hat{\beta}(v) - \beta(v) \right), \sqrt{n} \left(\hat{\beta}(w) - \beta(w) \right) \right\} = A(v)^{-1} B(v, w) A(w)^{-1}$$

where

$$B(v, w) = \mathbb{E}\{W_i(v) W_i(w) \{Y_i(v) - X_i^T \beta(v)\} \{Y_i(w) - X_i^T \beta(w)\} X_i X_i^T\}$$

Joint distribution of test statistic image $T_{m_1}(v)$

- $T_{m_1}(v) = n \times \left(\hat{\beta}(v) - \beta(v) \right)^T A(v) B(v, v)^{-1} A(v) \left(\hat{\beta}(v) - \beta(v) \right)$
- So $T_{m_1}(v) = Z(v)^T Z(v)$, where

$$Z(v) = \sqrt{n} \times B(v, v)^{-1/2} A(v) \left(\hat{\beta}(v) - \beta(v) \right) \sim N(0, I_{m_1})$$

Choosing subject weights for the sPBJ

- Optimal weights are the inverse of the observation level variance
- Optimal weight: $W_i(v) \approx \Sigma_i(v, v)^{-1}$
- Good weight: W_i is inverse of in-scanner motion
- No weights: still good asymptotically

$Y_1(v)$		$\text{Cov}\{Y_1(v), Y_1(w) X_1\} = \Sigma_1(v, w)$
$Y_2(v)$		$\text{Cov}\{Y_2(v), Y_2(w) X_2\} = \Sigma_2(v, w)$
	$\vdots$	
$Y_n(v)$		$\text{Cov}\{Y_n(v), Y_n(w) X_n\} = \Sigma_n(v, w)$

Some computational practicalities

- Estimation in neuroimaging often occurs in parallel across all locations
- Statistical Inference accounts for correlation among voxels
- sPBJ uses bootstrapping to estimate the joint distribution. More on this in Guillaume et al., 2014 and Vandekar et al., 2019 (readings)

Statistical Inference

This is where the spatial aspect of the data is incorporated

Options for statistical inference

- Types of inference
 - Voxel-wise inference
 - Cluster extent inference (spatial extent inference)
 - Threshold-free cluster enhancement
- Methods of inference
 - Gaussian random field theory
 - Permutation testing
 - Bootstrapping

Notation

$T_{m_1}(\boldsymbol{v})$ - Test statistic image for test of $H_0(\boldsymbol{v}): \beta(\boldsymbol{v}) = 0$

Can be T, Z, F, or Chi-square

m_1 - dimension of $\beta(\boldsymbol{v})$ for all $\boldsymbol{v}$

$p(\boldsymbol{v})$ - p-value image computed from $T_{m_1}(\boldsymbol{v})$

Hypothesis Testing (Inference)

We define particular hypotheses of interest such as:

$$H_0(v): \beta(v) = 0$$

(no effect of diagnosis on brain activation at location v)

or (more specifically)

No effect of diagnosis on assumed brain response profile of brain activation to positive versus negative emotions during *Despicable Me* clip at location v .

Hypothesis Testing

However, hypotheses like:

$$H_0(v): \beta(v) = 0$$

Are actually much more complex than they look.

For example, are we asking:

$$H_0(v): \beta(v) = 0 \text{ for all } v = 1, \dots, V$$

Or

$$H_0(v_0): \beta(v_0) = 0 \text{ for a specific } v_0 = 1, \dots, V?$$

And what precisely are the alternatives we are interested in?

Hypothesis Testing: Voxel-wise inference

Suppose we wish to test $H_{0v}: \beta(v) = 0$ for each $v = 0, \dots, m$. This is an exceedingly common set of hypotheses to test.

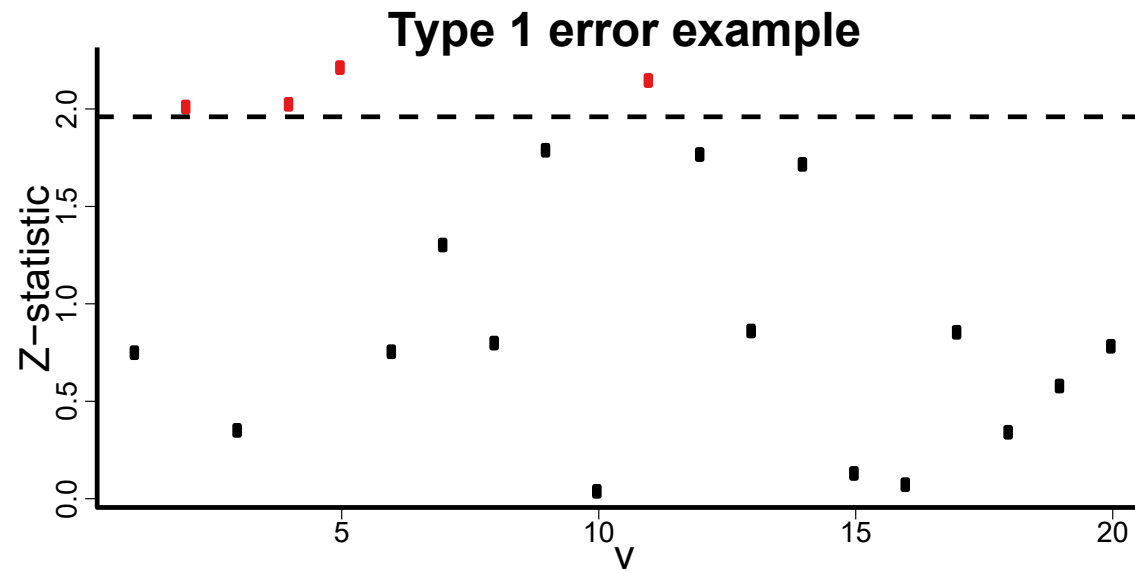
Usual process: define the type I error rate ($\alpha = 5\%$) and find a test statistic. If we have a good test statistic, then (under the null) approximately 5% of the time we will reject $H_{0v}: \beta(v) = 0$.

This is called **voxel-wise** inference – we are testing every voxel.

False positive rate in high dimensions

dataset-to-dataset variability

- For medical imaging data $m \approx 100,000$
- Performing the test of $H_0(v)$ for $v = 1, \dots, m$ leads to increased false positive rate



- $P(Z(v) > 1.96) = 0.05$
- $P(\cup_{v \in M_0} Z(v) > 1.96) = 1 - 0.95^{20} = 0.64$

Hypothesis Testing Errors

For each hypothesis test:

	$H_0(v)$ True	$H_0(v)$ False
Do Not Reject $H_0(v)$	Correct (probability $1-\alpha$)	Type II Error (probability β)
Reject $H_0(v)$	Type I Error (probability α)	Correct (probability $1-\beta$)

Across tests:

	# $H_0(v)$ True	# $H_0(v)$ False	Total #
Not Rejected $H_0(v)$	U	T	$V - R$
Rejected $H_0(v)$	W	S	R
	V_0	$V - V_0$	V

Hypothesis Testing Error Control

How do we control these errors? There are two common approaches:

	$\#H_0(v)\text{True}$	$\#H_0(v)\text{False}$	Total #
Not Rejected $H_0(v)$	U	T	$V - R$
Rejected $H_0(v)$	W	S	R
	V_0	$V - V_0$	V

Family-wise Error Rate (FWER)

$$FWER = P(W \geq 1)$$

False Discovery Rate (FDR)

$$FDR = E \left(\frac{W}{R} \mid R > 0 \right) P(R > 0)$$

Hypothesis Testing Error Control

How do we control these errors? There are two common approaches:

Family-wise Error Rate (FWER)

More conservative.

False Discovery Rate (FDR)

Less conservative.

Controlling the FWER - Bonferroni

The Bonferroni correction is the most common FWER correction.

It is notoriously conservative (high type II error rate).

Applying the Bonferroni correction is very simple: for each test H_{0v} , estimate a p-value p_v . Then, reject any hypothesis H_{0v} where

$$p_v \leq \frac{\alpha}{m}.$$

Controlling the FWER - Bonferroni

Reject any hypothesis H_{0v} where

$$p_v \leq \frac{\alpha}{m}.$$

But why? First, we need a lemma:

Lemma: Under $H_0: T \sim F_0$, we have that

$$p_t = P(T < t \mid T \sim F_0) \sim U(0,1)$$

$$\begin{aligned} \text{Proof: } P(p_t < u) &= P\{P(T < t \mid T \sim F_0) < u \mid t \sim F_0\} = P(F_0(t) < u \mid t \sim F_0) \\ &= P(t < F_0^{-1}(u) \mid t \sim F_0) = F_0\{F_0^{-1}(u)\} = u. \end{aligned}$$

Controlling the FWER - Bonferroni

Then, using Boole's inequality, and noting by the lemma that under H_{0v} we have $p_v \sim U(0,1)$,

$$FWER = P\left(\cup_{v=1}^V \left\{p_v \leq \frac{\alpha}{m}\right\}\right) \leq \sum_{v=1}^V P\left(p_v \leq \frac{\alpha}{m}\right) \leq \frac{m_0 \alpha}{m} \leq \alpha.$$

Note that this argument did not depend on how many hypotheses were true, nor did it depend on the distribution of the test statistics; in particular, it did not depend on any correlation between the test statistics.

Controlling the FWER – Boole's

Another relatively common technique for controlling the FWER is referred to as Holm's procedure.

- Order the p-values such that $p_{(1)} \leq \dots \leq p_{(m)}$.
- Adjust the p-values:

$$p_j^* = \min\{(m - j + 1)p_j, 1\}$$

Which is equivalent to finding the largest k such that

$$p_{(k)} > \frac{\alpha}{m - k + 1}$$

And rejecting $H_{0(1)} \leq \dots \leq H_{0(k-1)}$ but not $H_{0(k)} \leq \dots \leq H_{0(m)}$.

Note that the Holm procedure also controls the FWER under arbitrary dependence, invoking Boole's inequality.

Controlling the FDR

- Controlling the FDR results in more liberal inference, and higher statistical power, at the cost of FWER.
- There are several common methods for controlling the FDR, but the most common is called the Benjamini-Hochberg procedure.
 - Controls the FDR under independence.
 - Controls the FDR under positive dependence.
- A modification of this procedure is also available for more general dependencies and is known as the Benjamini-Yekutieli procedure.
 - It is known to provide much less power, and so is less common.

Controlling the FDR - BH

To control the FDR, Benjamini and Hochberg (1995) suggested that we can control the FDR at level δ by:

- Ordering the p-values such that $p_{(1)} \leq \dots \leq p_{(m)}$.
- Finding the largest k such that

$$p_{(k)} \leq \frac{k}{m} \delta$$

And rejecting $H_{0(1)} \leq \dots \leq H_{0(k)}$ but not $H_{0(k)} \leq \dots \leq H_{0(m)}$.

These methods ignore joint distribution. What else can we do?

- Another (less common) approach was pioneered by Westfall and Young (1993) and is based on permutation:
- We permute the rows of the design matrix (X_i) repeatedly and generate the distribution of the maximum test statistic (or min p-value).
- Then we compare this empirical distribution of the maximum test statistic with the observed distribution of the test statistics.

Many spatial FDR methods

J. R. Statist. Soc. B (2015)
77, Part 1, pp. 59–83

False discovery control in large-scale spatial multiple testing

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False Discovery Rates for Spatial Signals

HELLER

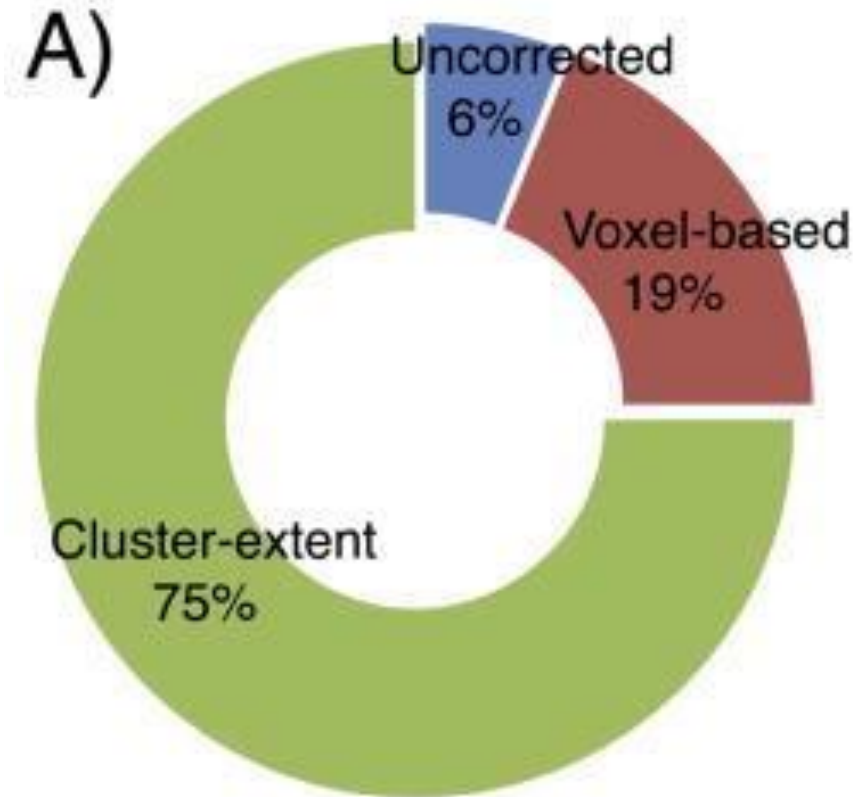
ng for the presence of signal in spatial data can involve numerous locations. Traditionally, each location is presence, but then the findings are reported in terms of clusters of nearby locations. This is an indication that the clusters rather than individual locations. The investigator may know a priori these more natural units or suggest testing these cluster units rather than individual locations, thus increasing the signal-to-noise ratio within cing the number of hypothesis tests conducted. Because the signal may be absent from part of each cluster, ing a signal if the signal is present somewhere within the cluster. We suggest controlling the false discovery ie expected proportion of clusters rejected erroneously out of all clusters rejected) or its extension to general ce a powerful two-stage testing procedure and show that it controls the WFDR. Once the cluster discoveries cleaning” locations in which the signal is absent. For this purpose, we develop a hierarchical testing procedure its locations within rejected clusters. We show formally that this procedure controls the desired location error cture that this is also so for realistic settings by extensive simulations. We discuss an application to functional his research and demonstrate the advantages of the proposed methodology on an example.

magnetic resonance imaging; Hierarchical testing; Multiple testing; Signal detection; Weighted testing proce-

; Power; Signal detection; Wavelets.

Cluster extent inference most widely used approach in medical imaging

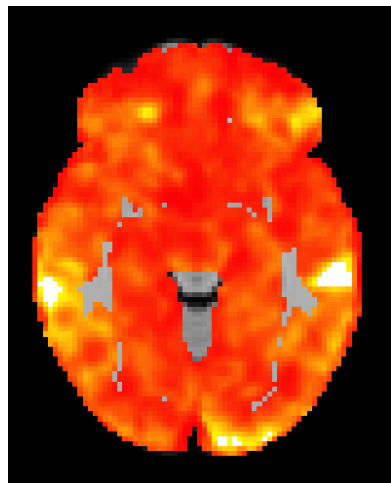
- Multiple comparisons correction
 - FWER
 - FDR
- Voxel-wise correction
- Cluster extent inference
- Other approaches
 - TFCE
 - Cluster-mass



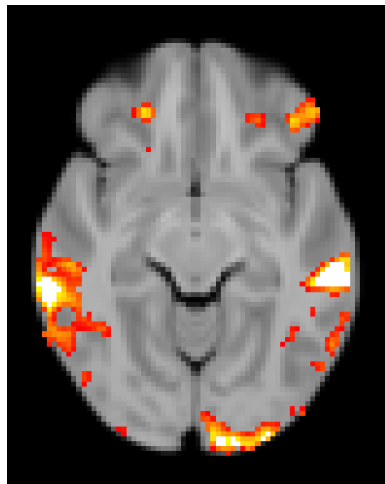
Woo, Krishnan, & Wager, *Neuroimage*, 2014

Cluster extent inference procedure

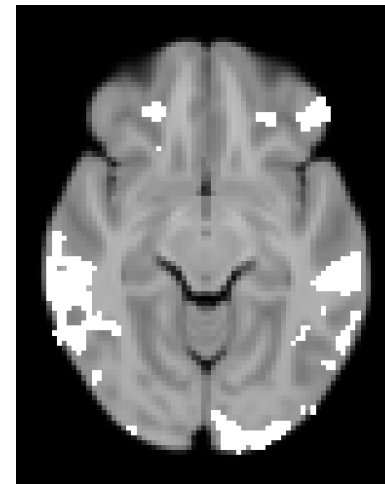
$T_{m_1}(v)$ - Chi-squared
 statistic image for test
 of $H_0(v): \beta(v) = 0$
 $p(v)$ - p-value image
 computed from $T_{m_1}(v)$
 m_1 - dimension of $\beta(v)$



Threshold
stat image
(CFT)



Binarize
thresholded
stat image



Compute
cluster
statistics

Cluster	Num voxels	Vol (mm ³)
1	157	628
2	67	268
3	57	228
⋮	⋮	⋮
16	1	4

Statistic image, $T_{m_1}(v)$,
for test of: $H_0(v): \beta(v) = 0$

- Image viewed as a “random field”
- Choose **cluster forming threshold (CFT)** $p(v) < 0.001$
(i.e. $T_1(v) > 10.82$)

$T_{m_1}(v)$ - Chi-squared
statistic image for test
of $H_0(v): \beta(v) = 0$
 $p(v)$ - p-value image
computed from $T_{m_1}(v)$
 m_1 - dimension of $\beta(v)$

Cluster extent inference p-values

New set of hypotheses that are cluster specific:

- Let $E(c)$ denote the cluster extents for $c = 1, \dots, C$, where c indexes clusters in decreasing size
- $H_0(c): E(c) =_d E_0(c)$, where $E_0(c)$ assumes $H_0(v): \beta(v) = 0 \forall v$
- Compute cluster adjusted p-value

$$p(c) = P(E_0(1) > e(c)),$$

- where $e(c)$ is the observed cluster extent for cluster c
- **Computing $p(c)$ requires an estimate of the joint distribution of $T_{m_1}(v)$**

Spatial extent inference interpretation

- $p(c)$ is the probability of observing a cluster size at least as large as cluster c if the image is not associated with the covariate

Three approaches to CEI

1. Gaussian random field
2. Permutation method – randomise from FSL¹
3. Parametric bootstrap joint (PBJ) testing procedure²

¹Winkler et al., *Neuroimage*, 2014

²Vandekar, *Arxiv*, 2018

Gaussian Random Field Theory

- This is a fancy term describing the fact that the data across locations are correlated, and we can make assumptions about them.
- Most commonly, we make the assumption that the test statistics of interest are normally distributed but correlated across locations, which is called a Gaussian Random Field (GRF).
- Usually, we assume that the smoothness in the image is the same across locations – this can get us into trouble.
- The effective number of observations is less than the number of voxels, and this can be expressed as the number of “resels” (Worsley et al., 1992). This allow us to threshold more liberally while adjusting for multiple comparisons.

Gaussian Random Field Theory

- GRF approximations require a parameter describing spatial smoothness. This can be:
 - Externally entered, if known a priori.
 - Estimated from the data.
- GRF assumptions can give us the null distribution of the maximum test statistic, which we can use to do voxel-wise testing.
 - It's fast and gives a threshold for p-values that controls FWER.
 - But it hinges on the assumptions.
- What else can it give us?

References

- [Friston et al. \(1994\) *Assessing the significance of focal activations using their spatial extent*](#)
- [Brett, Johnsrude & Owen \(2002\) *Introduction to Random Field Theory*](#)
- [Worsley et al. \(1992\) *Three-dimensional statistical analysis for CBF activation studies*](#)
- [Woo, Krishnan & Wager \(2014\) *Cluster-extent based thresholding in fMRI analyses*](#)
- [Eklund, Nichols & Knutsson \(2016\) *Cluster failure: Why fMRI inferences...*](#) * **Nonstationary RFT:** [Hayasaka et al. \(2004\)](#)
- **Cluster-Mass RFT:** [Zhang et al. \(2009\)](#)
- **Threshold-Free Cluster Enhancement (TFCE):** [Smith & Nichols \(2009\)](#)
- Peaks as test statistics, EC gives expected number of maxima.
- [Cao & Worsley \(1999\) *The distribution of the maximum of Gaussian random fields on 6D correlation matrices*](#)
- [Davenport & Nichols \(2020\) *Selective peak inference...*](#)
- [Davenport et al. \(2023\) *Robust FWER control in neuroimaging using RFT*](#)

Problem: inflated FWER in neuroimaging

- Standard mass-univariate tools have inflated FWER
- **Unrealistic assumptions about covariance structure**

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Cluster failure: Why fMRI inferences for spatial extent have inflated false-positive rates

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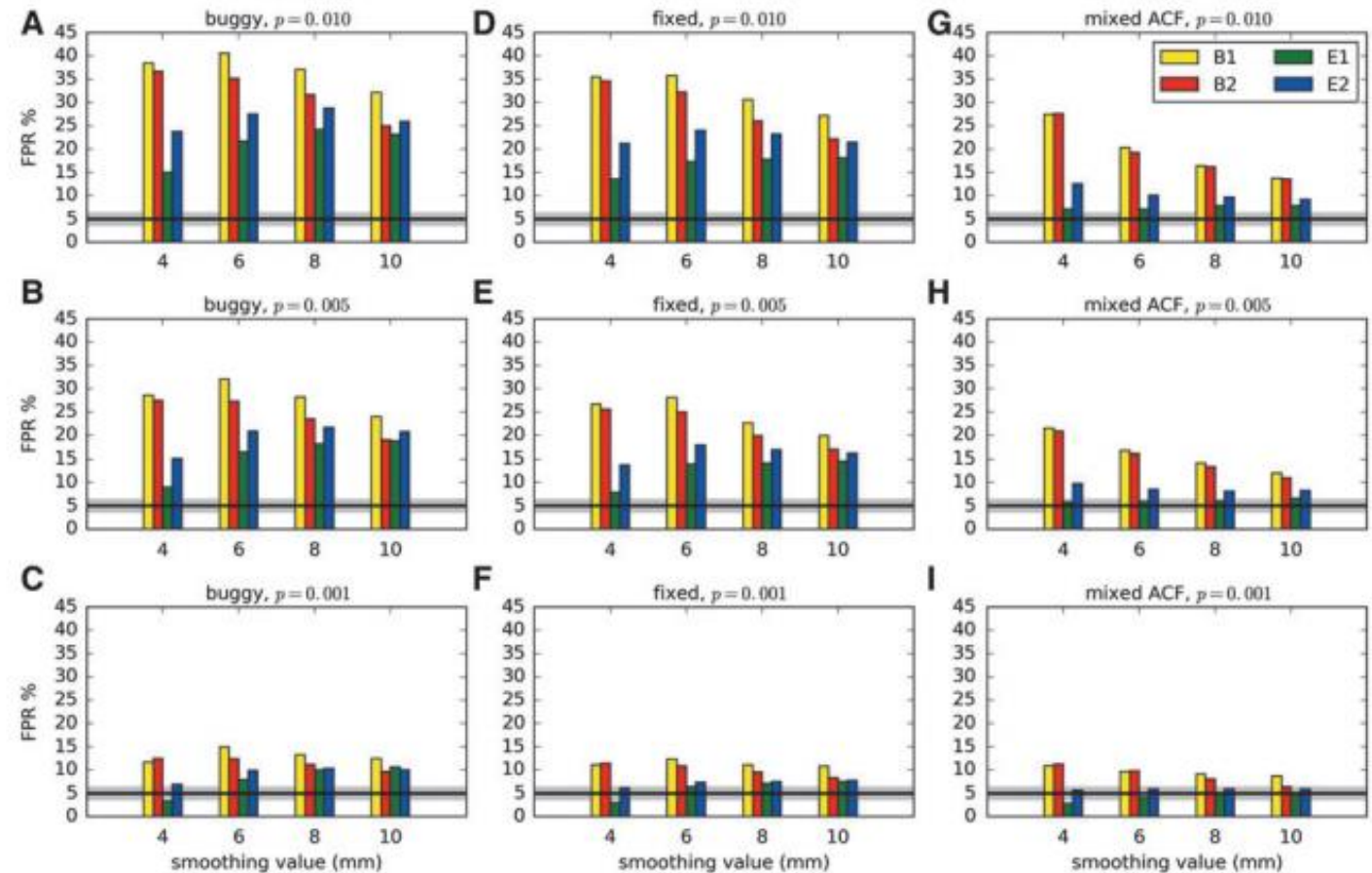
Edited by Emery N. Brown, Massachusetts General Hospital, Boston, MA, and approved May 17, 2016 (received for review February 12, 2016)

fMRI brain activations associated with tasks/activities/thoughts was found to deliver inflated false-positive rates.

Unrealistic assumptions in CEI

- Classical methods for SEI rely on smooth Gaussian random field approximations
- GRF approximations only work under very restrictive assumptions
- Many papers on this:

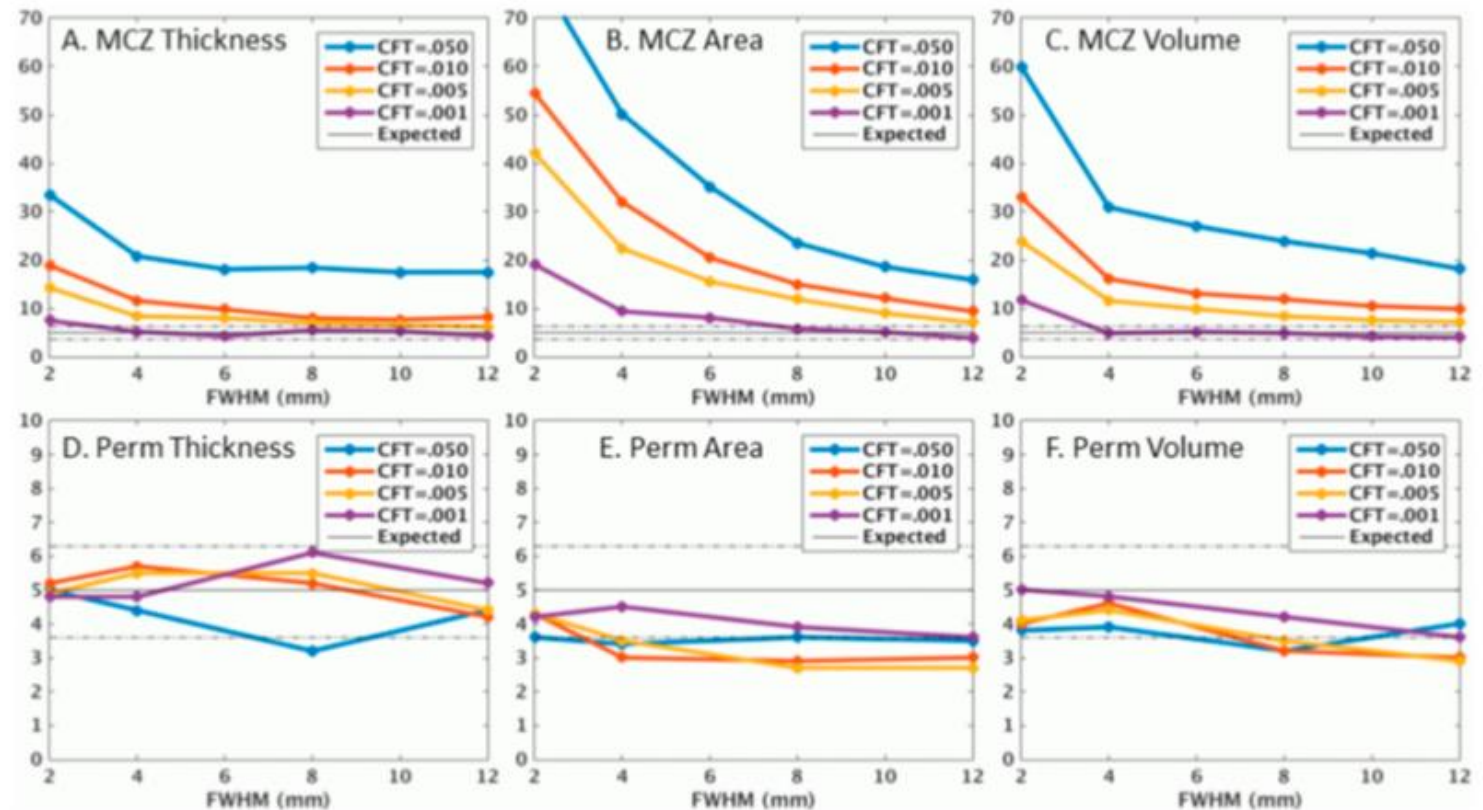
Silver 2011, Woo 2014, Eklund 2016, Flandin 2016, Slotnick 2017, Cox 2017, Kessler 2017, Mueller 2017, Greve 2018



Cox et al., *Brain Connectivity*, 2017

Permutation tests work quite well

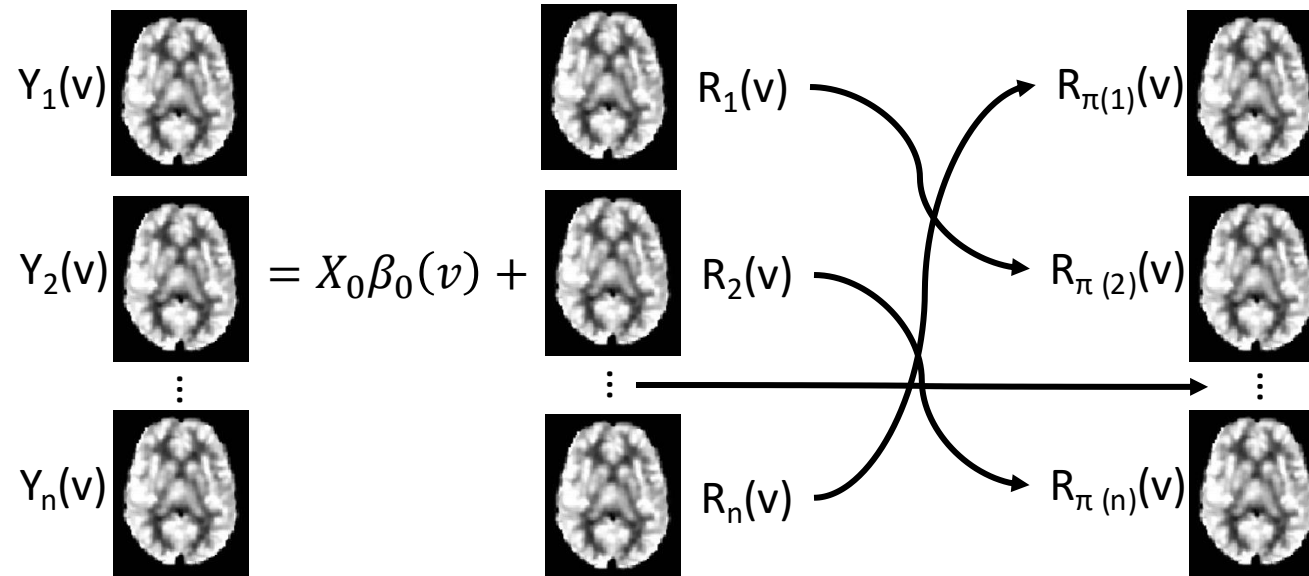
- Permutation testing maintains nominal FWER
- Limitation: requires exchangeability – cannot accommodate multilevel models or heteroskedasticity
- Our methodological goal is to develop a inference procedure appropriate for multilevel models



Greve & Fischl, *Neuroimage*, 2018

Permutation CEI procedure

Step 1: Get residuals from null model



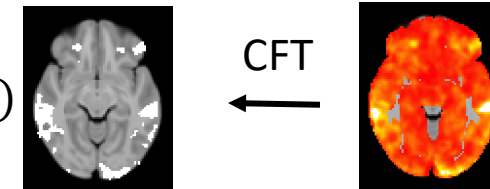
Steps 2 to P+1: Fit model to randomly permuted residuals

$$= X_0\beta_0(v) + X_1\beta_{1p}(v) + \epsilon_p(v)$$

$$H_0(v): \beta_{1p}(v) = 0$$

Step P+2: Compute adjusted p-values

$$p_{\text{adj}}(c) = P^{-1} \# \{E_p(1) > e(c)\} \leftarrow E_p(1)$$



Permutation procedure with covariates

- Randomise (FSL permutation procedure) uses the Freedman-Lane method
- They evaluated many in their paper

Table 2

A number of methods are available to obtain parameter estimates and construct a reference distribution in the presence of nuisance variables.

Method	Model
Draper-Stoneman ^a	$Y = PX\beta + Z\gamma + \epsilon$
Still-White ^b	$PR_Z Y = X\beta + \epsilon$
Freedman-Lane ^c	$(PR_Z + H_Z)Y = X\beta + Z\gamma + \epsilon$
Manly ^d	$PY = X\beta + Z\gamma + \epsilon$
ter Braak ^e	$(PR_M + H_M)Y = X\beta + Z\gamma + \epsilon$
Kennedy ^f	$PR_Z Y = R_Z X\beta + \epsilon$
Huh-Jhun ^g	$PQ'R_Z Y = Q'R_Z X\beta + \epsilon$
Smith ^h	$Y = PR_Z X\beta + Z\gamma + \epsilon$
Parametric ⁱ	$Y = X\beta + Z\gamma + \epsilon, \epsilon \sim N(0, \sigma^2 I)$

Winkler et al., *Neuroimage*, 2017

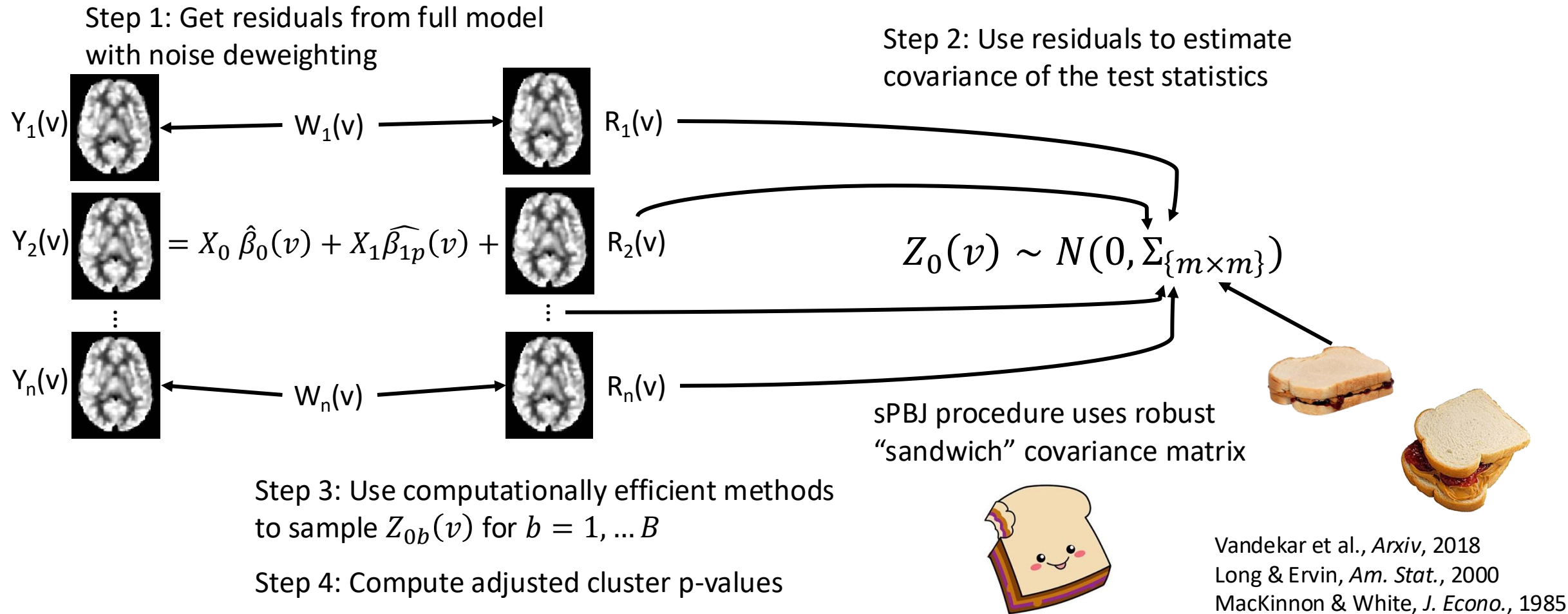
Freedman-Lane procedure

1. Regress $\mathbf{Y}$ against the full model that contains both the effects of interest and the nuisance variables, i.e. $\mathbf{Y} = \mathbf{X}\beta + \mathbf{Z}\gamma + \epsilon$. Use the estimated parameters $\hat{\beta}$ to compute the statistic of interest, and call this statistic T_0 .
2. Regress $\mathbf{Y}$ against a reduced model that contains only the nuisance effects, i.e. $\mathbf{Y} = \mathbf{Z}\gamma + \epsilon_Z$, obtaining estimated parameters $\hat{\gamma}$ and estimated residuals $\hat{\epsilon}_Z$.
3. Compute a set of permuted data $\mathbf{Y}_j^*$. This is done by pre-multiplying the residuals from the reduced model produced in the previous step, $\hat{\epsilon}_Z$, by a permutation matrix, $\mathbf{P}_j$, then adding back the estimated nuisance effects, i.e. $\mathbf{Y}_j^* = \mathbf{P}_j\hat{\epsilon}_Z + \mathbf{Z}\hat{\gamma}$.
4. Regress the permuted data $\mathbf{Y}_j^*$ against the full model, i.e. $\mathbf{Y}_j^* = \mathbf{X}\beta + \mathbf{Z}\gamma + \epsilon$, and use the estimated $\hat{\beta}_j^*$ to compute the statistic of interest. Call this statistic T_j^* .
5. Repeat Steps 2–4 many times to build the reference distribution of T^* under the null hypothesis.
6. Count how many times T_j^* was found to be equal to or larger than T_0 , and divide the count by the number of permutations; the result is the p-value.

Permutation SEI procedure pros and cons

- Pros:
 - It is more robust than GRF based methods
 - It is easily implemented for linear models with the “randomise” FSL function
- Cons
 - It is not appropriate for multilevel models e.g. fMRI
 - Requires exchangeability assumption i.e. $\text{Var}\{\hat{\beta}_i(v), \hat{\beta}_i(w)\} = \Sigma(v, w)$
 - It can take over a day to run

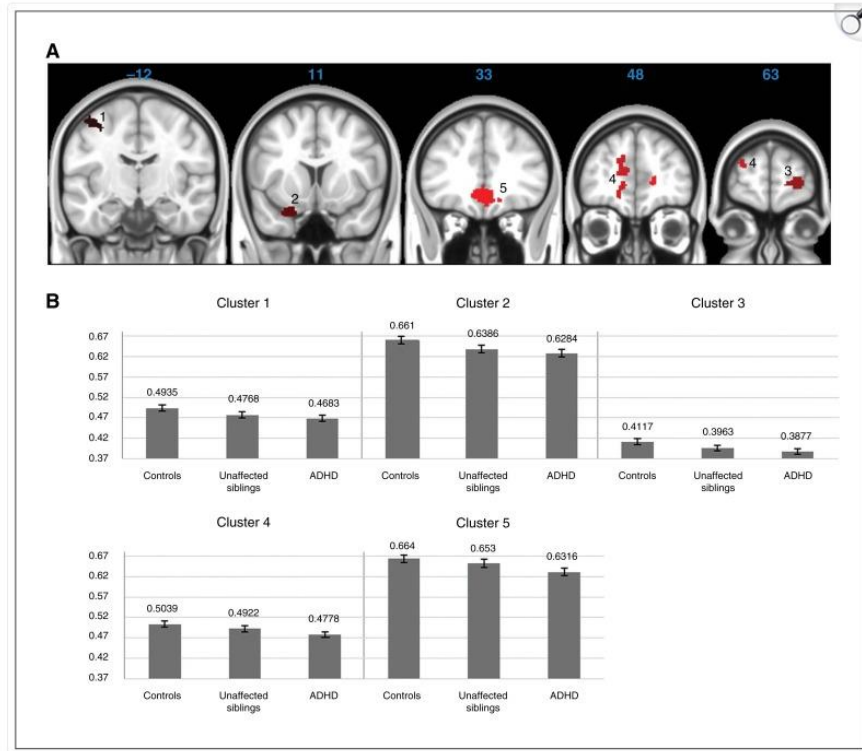
Semi-PBJ (sPBJ) SEI procedure



Typical statistical reporting

Bratlen et al., *J Psychiatry Neurosci*, 2015

Voxel-based morphometry analysis reveals frontal brain differences in participants with ADHD and their unaffected siblings



- They compared structural differences at the voxel-level between ADHD, sibling, and HC
- Questions:
 - What is the interpretation of the results?
 - What kind of test was performed at each voxel? What is the contrast?
 - What kind of spatial inference was used?
- **Methods:** We considered differences to be significant if they survived cluster-mass thresholding with the *easythresh* option in FSL (www.fmrib.ox.ac.uk/fsl), using an initial cluster forming threshold of $z > 3.1$. Subsequently, we estimated each cluster's significance level based on Gaussian random field theory, and those clusters surviving a family-wise error (FWE)-corrected significance threshold of $p < 0.05$ showing volume differences > 0.1 mL were reported.

• Results:

(A) Whole-brain significant clusters for case-control differences. Five clusters were identified: cluster 1 = precentral gyrus; cluster 2 = orbitofrontal cortex; cluster 3 = frontal pole; cluster 4 = paracingulate and cingulate cortices, frontal pole; and cluster 5 = medial frontal, paracingulate, cingulate and subcallosal cortices. (B) Mean voxel differences for the identified cluster between participants with attention-deficit/hyperactivity disorder (ADHD; $n = 307$), their unaffected siblings ($n = 169$) and typically developing controls ($n = 196$).

Bratlen et al., *J Psychiatry Neurosci*, 2015

- Cluster summary table

Cluster [*]	No. of voxels	MNI coordinates (x, y, z) [†]	Best z value	Side of the brain	Area [‡]
Cl 1	157	−40, −6, 56	−3.96	L	Precentral gyrus
Cl 2	244	−26, 16, −24	−4.43	L	Orbitofrontal cortex
Cl 3	250	28, 70, −2	−4.17	R	Frontal pole
Cl 4	518	−14, 52, 14	−4.43	L	Paracingulate cortex, cingulate cortex, frontal pole
Cl 5	667	2, 22, −2	−3.79	L, R	Medial frontal, paracingulate, cingulate and subcallosal cortices

Posthoc comparisons

- Analyses in significant clusters.
- What is different from the C-A p-values shown here?

Mean voxel values comparisons for participants with ADHD, their unaffected siblings and controls

Cluster	C-U <i>p</i> value	R^2_{-}	U-A <i>p</i> value	R^2_{-}	C-A <i>p</i> value	R^2_{-}	Group; mean $\pm$ SD		
							Controls	Unaffected siblings	ADHD
Cluster 1	0.016	0.016	0.18	0.003	< 0.001	0.034	0.494 $\pm$ 0.0652	0.477 $\pm$ 0.0686	0.468 $\pm$ 0.0626
Cluster 2	0.003	0.027	0.10	0.005	< 0.001	0.049	0.661 $\pm$ 0.0723	0.639 $\pm$ 0.0686	0.628 $\pm$ 0.0723
Cluster 3	0.007	0.022	0.09	0.006	< 0.001	0.043	0.412 $\pm$ 0.0573	0.396 $\pm$ 0.0568	0.388 $\pm$ 0.0590
Cluster 4	0.034	0.041	0.007	0.016	< 0.001	0.052	0.504 $\pm$ 0.0544	0.492 $\pm$ 0.0576	0.478 $\pm$ 0.0562
Cluster 5	0.21	0.004	0.004	0.016	< 0.001	0.037	0.664 $\pm$ 0.0823	0.653 $\pm$ 0.0833	0.632 $\pm$ 0.0762

ADHD = attention-deficit/hyperactivity disorder; C-A = mean volume differences between controls and ADHD; C-U = mean volume differences between controls and unaffected siblings; U-A = mean volume differences between unaffected siblings and ADHD; SD = standard deviation.

*Effect sizes (R^2) are based on mean cluster comparisons using robust cluster regression in Stata software after regressing out age, age², sex and scanner site.

Loitfelder et al., *PLOS One*, 2014

Brain Activity Changes in Cognitive Networks in Relapsing-Remitting Multiple Sclerosis – Insights from a Longitudinal fMRI Study

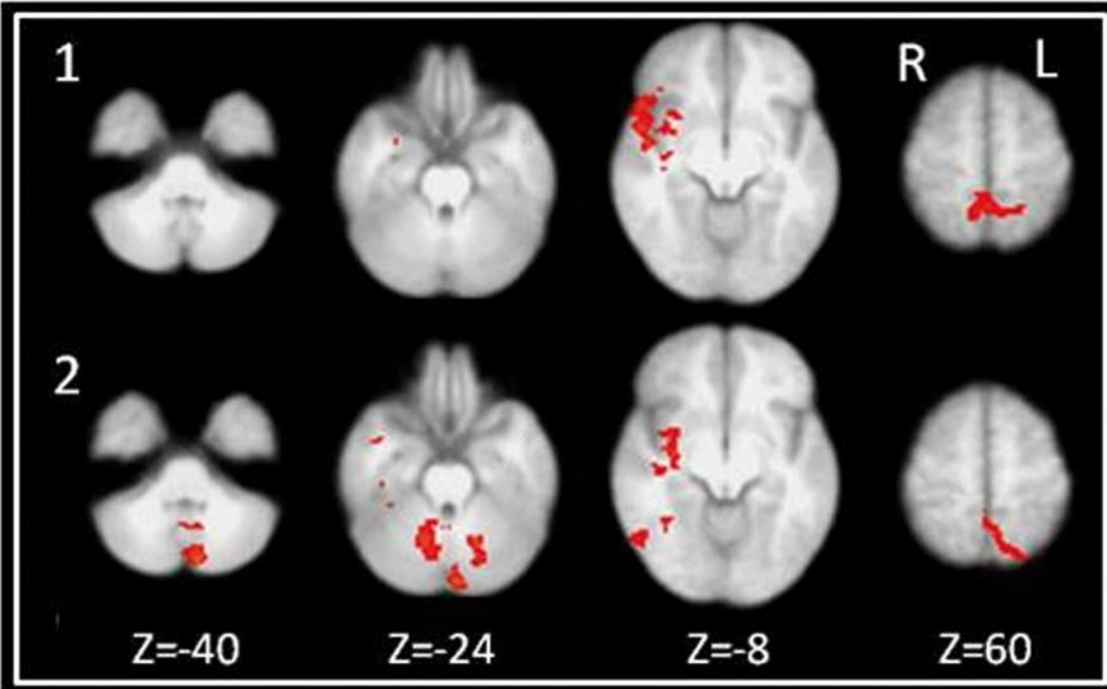


Figure 2. Areas with increased brain activation in MS patients vs. controls.

Clusters of significant activation difference (mixed effects higher level analyses; $Z > 2.3$; corrected cluster significance threshold $p = 0.05$) in contrasts for Go/-noGo task vs. rest for MS patients compared to controls at baseline (1) and follow-up (2).

- They compared functional changes at the voxel-level between MS and HC during a Go/No-Go task
- Questions:
 - What model was used?
 - What kind of test was performed at each voxel? What is the contrast?
 - What kind of spatial inference was used?
- **Methods:** Higher-level analysis was done using FLAME stage 1 (FMRIB's Local Analysis of Mixed Effects). Z (Gaussianised T/F) statistic images were thresholded using clusters determined by $Z > 2.3$ and a (corrected) cluster significance threshold of $p = 0.05$ (for further details see [\[4\]](#)). At higher level, contrasts were calculated within-groups over time and across groups over time.
- **Results:** Patients demonstrated increased activation compared to HC in the insular cortex and precuneus at BL. At FU, they additionally activated the posterior cingulate cortex (PCC), the cerebellum (left crus II, right lobule VI), and the lateral occipital cortex (superior and inferior division, Figure 2, Table 3).

Loitfelder et al., *PLOS One*, 2014

Cluster Statistics for clusters of significant differences (coordinates, maximum z-score, cluster size) between MS patients vs. healthy controls for Go-/noGo conditions vs. rest at baseline and follow-up.

Region	L/R	x	y	z	Z-max	cluster size
<i>Baseline</i>						
insular cortex	R	52	4	−6	3.46	1048
precuneus	R	0	−40	52	3.39	574
<i>Follow-up</i>						
insular cortex	R	34	−4	−12	3.69	570
cerebellum (crus II)	L	−8	−80	−40	3.73	508
PCC	L	−2	−36	48	3.78	483
lateral occipital cortex, superior division	R	42	−88	10	3.52	467
lateral occipital cortex, inferior division	R	56	−62	−6	3.36	409
cerebellum, area VI	R	14	−60	−24	3.39	376

doi:10.1371/journal.pone.0093715.t003

Confidence sets

Confidence set inference

Extras after this

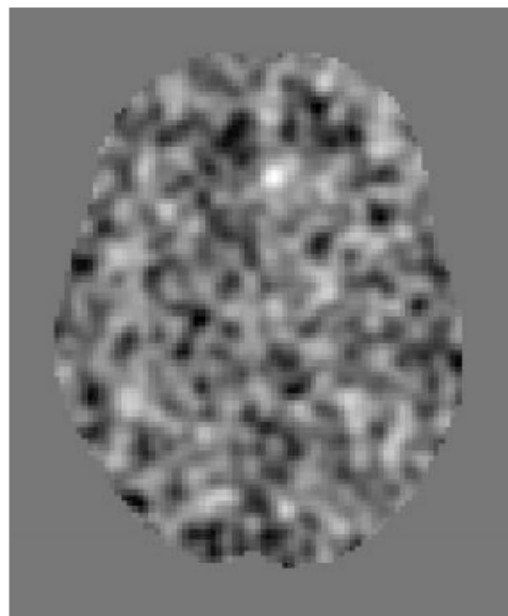
FSL slides on GRF and permutation



Distribution of Max Cluster Size

So, just as was the case for the z -values, we now have a distribution $f(R)$ that allows us to calculate a Family Wise threshold $u(R)$ pertaining to cluster size.

$f(R)$ depends crucially on the initial “cluster-forming” threshold?

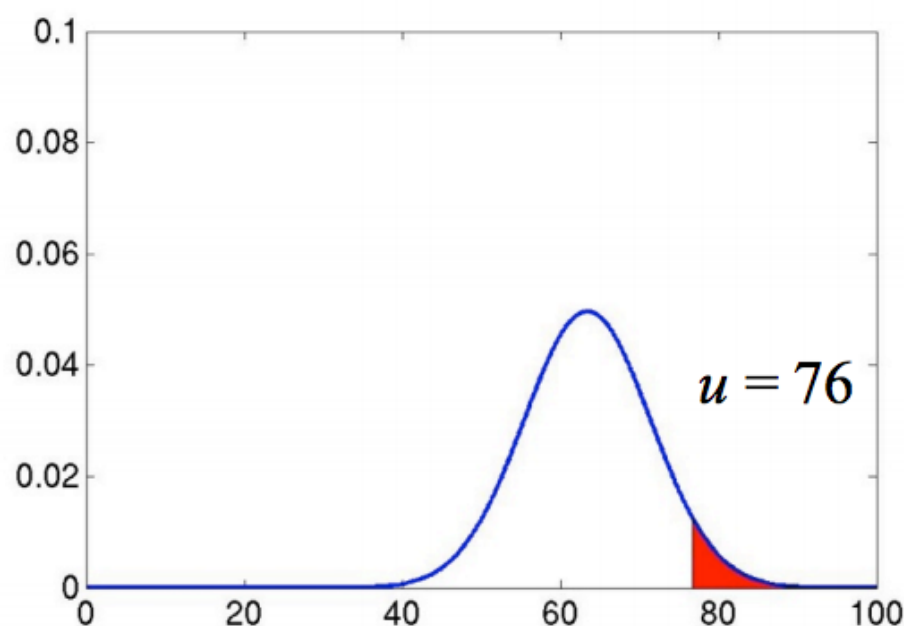


$$z = 2.3$$

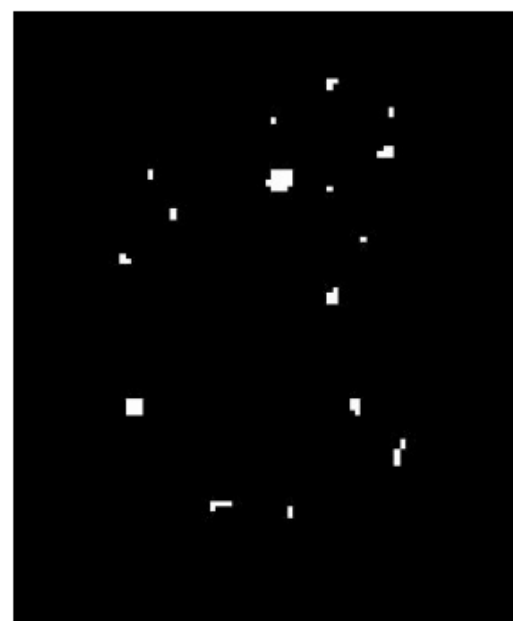


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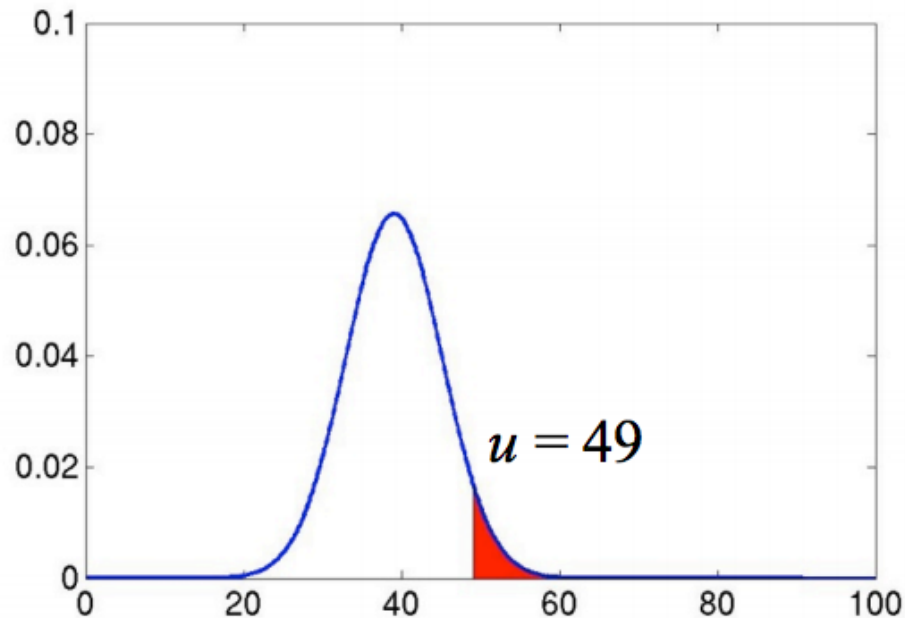


$z = 2.3$

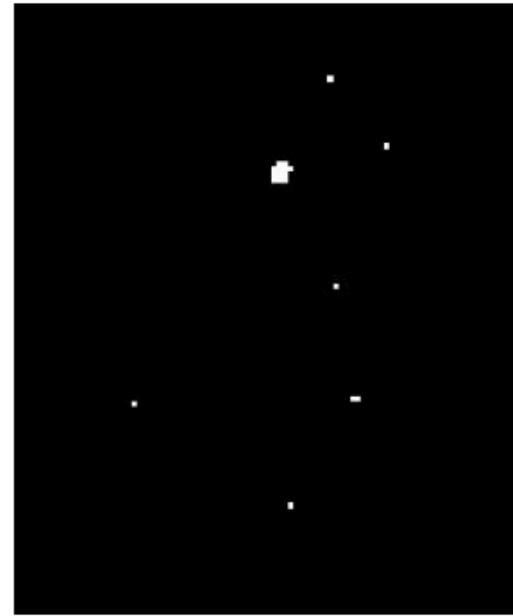


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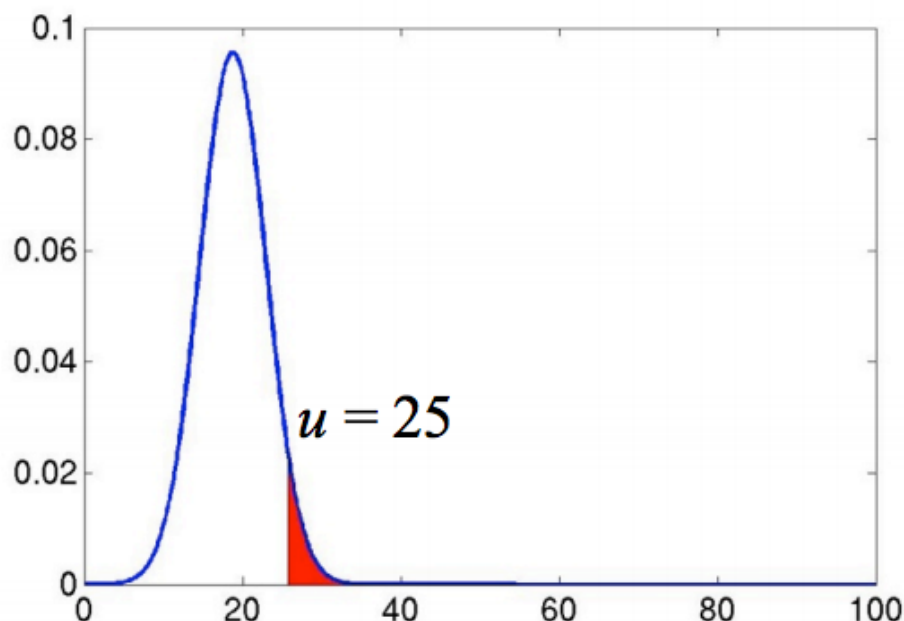


$z = 2.7$

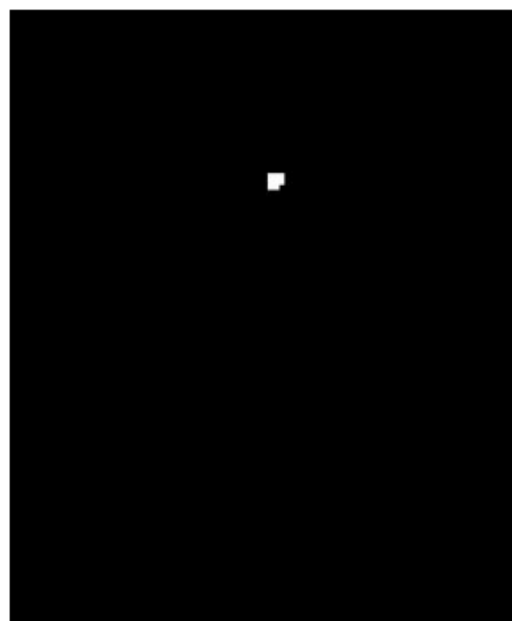


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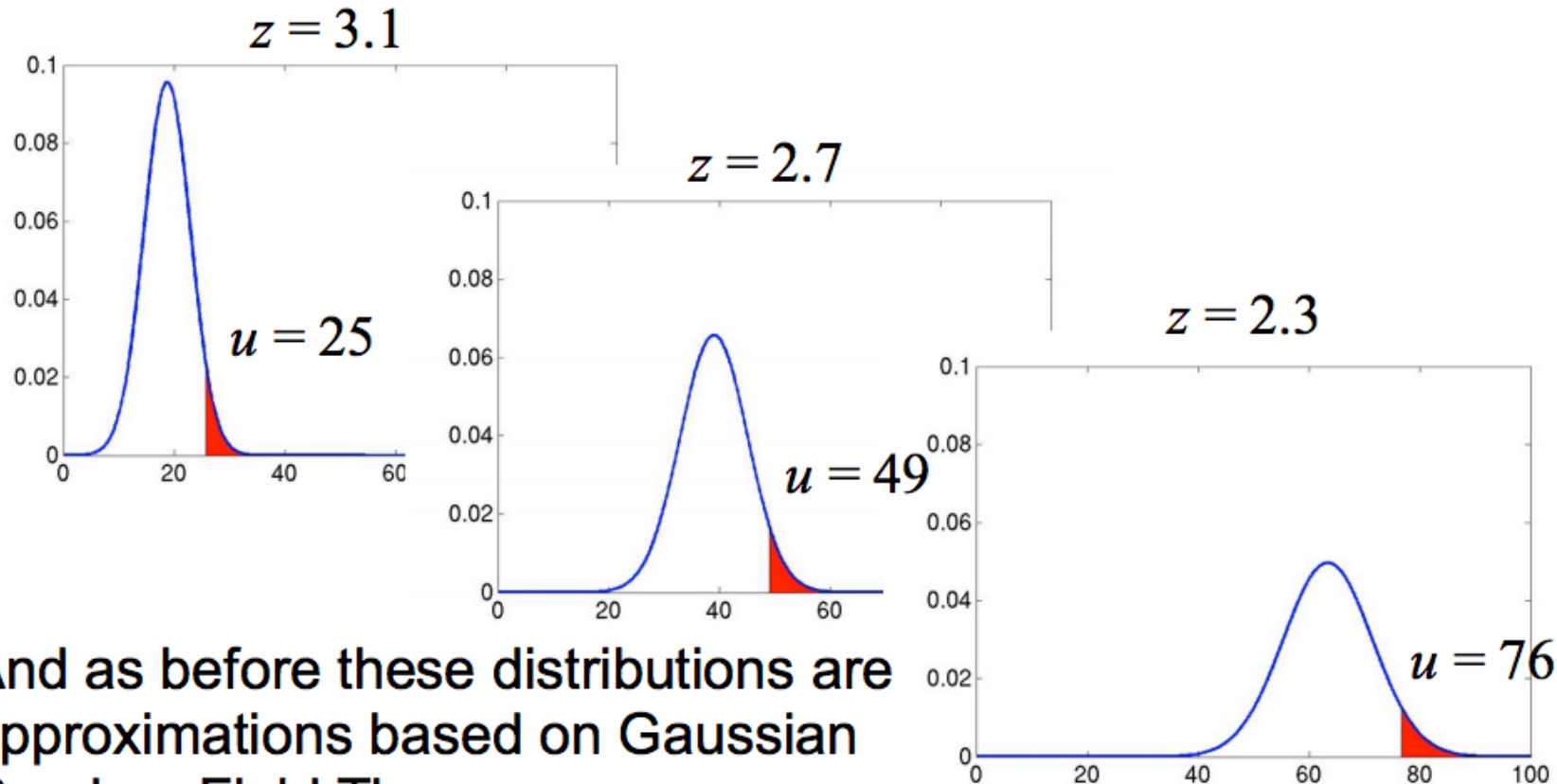


$z = 3.1$



Distribution of Max Cluster Size

Hence the distribution for the cluster size should really be written $f(R,z)$ and the same for $u(R,z)$

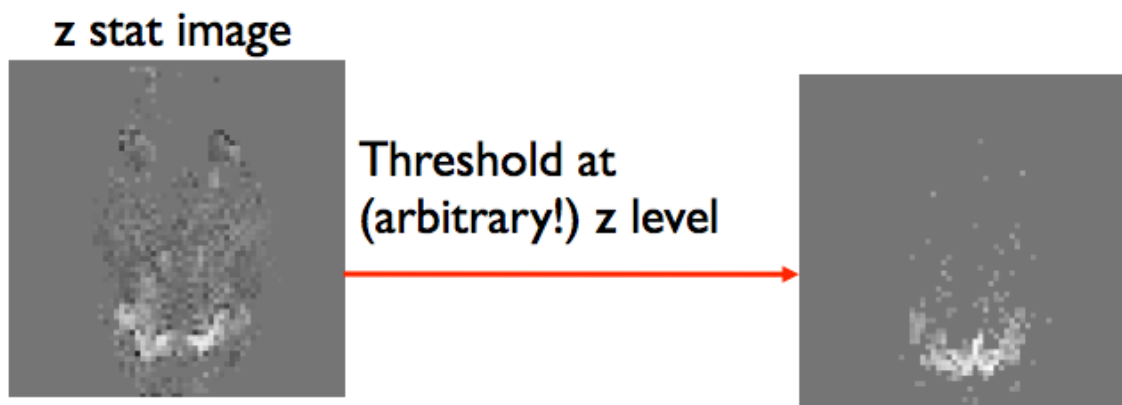


And as before these distributions are approximations based on Gaussian Random Field Theory.



Clustering cookbook

Instead of resel-based correction, we can do clustering:





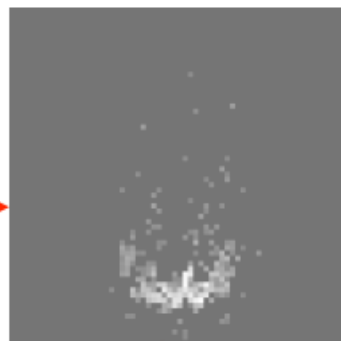
Clustering cookbook

Instead of resel-based correction, we can do clustering

z stat image



Threshold at
(arbitrary!) z level



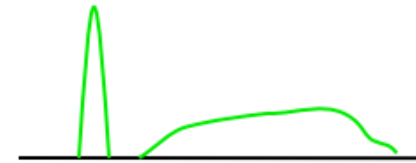
Form clusters from surviving voxels.
Calculate the size threshold $u(R,z)$.
Any cluster larger than u “survives” and we reject
the null-hypothesis for that.





How do we choose the (arbitrary!) z-threshold?

This is arbitrary and a trade-off

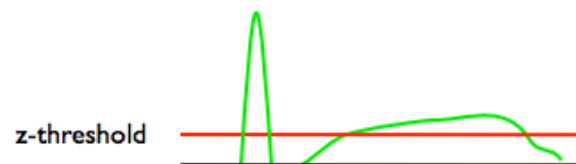




How do we choose the (arbitrary!) z-threshold?

This is arbitrary and a trade-off

1. **Low threshold** - can violate RFT assumptions, but can detect clusters with large spatial extent and low z

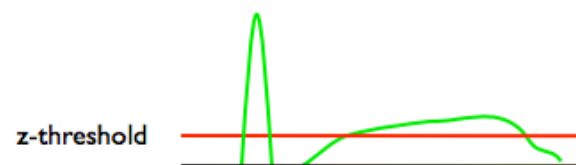




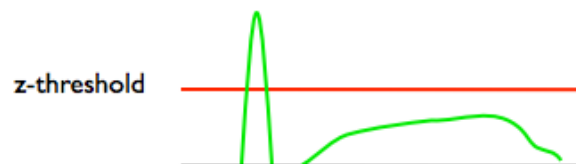
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2. **High threshold** - gives more power to clusters with small spatial extent and high z

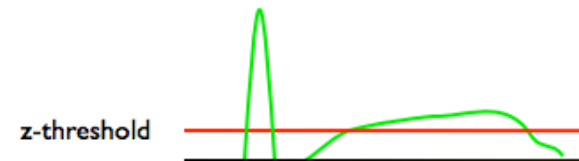




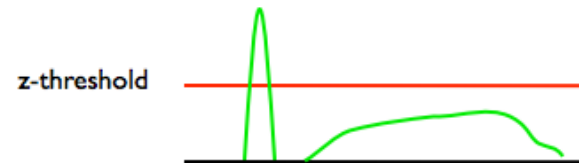
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Tends to be more sensitive than voxel-wise corrected testing

Results depend on extent of spatial smoothing in pre-processing



GRF for cluster-wise tests

- Needs a null-hypothesis, a test-statistic and an initial cluster forming threshold.
- Calculates a (size) threshold based on number of RESELS and initial (z) threshold
- + Gives a (size) threshold such that the family-wise error is controlled.
- + Calculates that threshold very fast.
- Hinges on strict assumptions about the distribution of the data ($\sim N(0, \Lambda)$)
- Inference pertains to entire cluster
- Initial threshold is arbitrary



Randomise, slow and reliable.

Impressive as the GRF based thresholding is, there are situations where we can't use it.



randomise always delivers

We may want to use a test-statistic for which the distribution is unknown.

Example: The “TFC enhanced” t -statistic. (A highly non-linear spatial filter)

Our data may not be normally distributed. Then our t -values will, paradoxically, not be t -distributed. (what are the chances...)

Example: VBM-style data (data whose value is probability of a certain tissue-type)

We want to restrict our analysis to a particular (small and irregularly shaped) sulcus to increase our sensitivity when we have a prior spatial hypothesis.



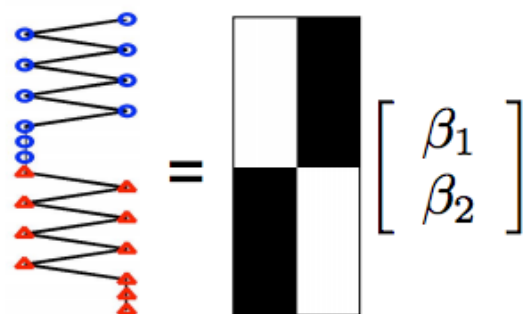
Example: VBM-style analysis

- Our data is segmented grey matter maps
- A voxel is either grey matter, or not.

Group #1
(Oxford students)

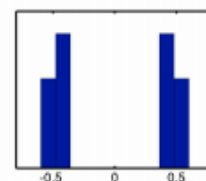


Group #2
(Train spotters)



$$\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} \text{ Ok!}$$

hist(e)



$\sim N?$

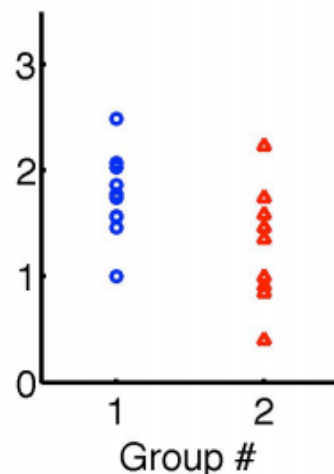




Oh dear! What now?

- We could use Monte-Carlo to simulate the distributions.
As I did for these slides.
 - But, hinges on lots of assumptions about the data
- We could **permute** the data itself.

We have performed an
experiment



And calculated a
statistic, e.g. a t -value

$$t = 2.27$$

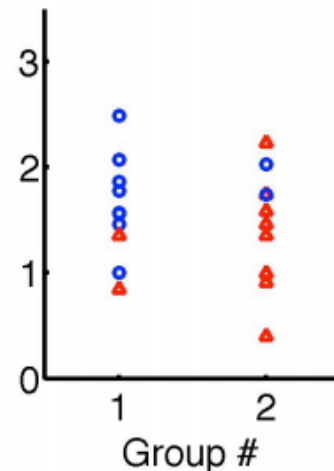
If the null-hypothesis is true, there is no difference between the groups. That means we should be able to “re-label” the individual points without changing anything.



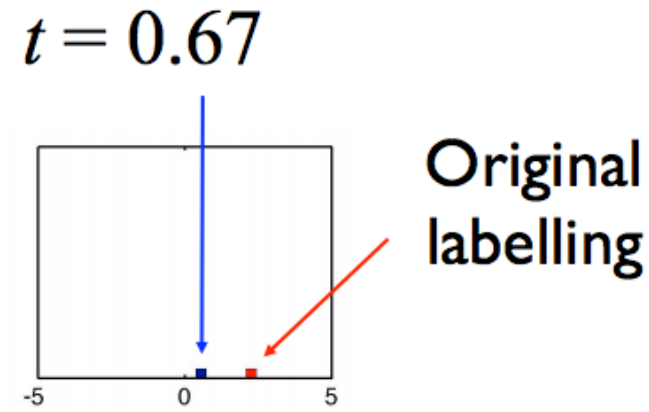
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One re-labelling



t -value after re-labelling



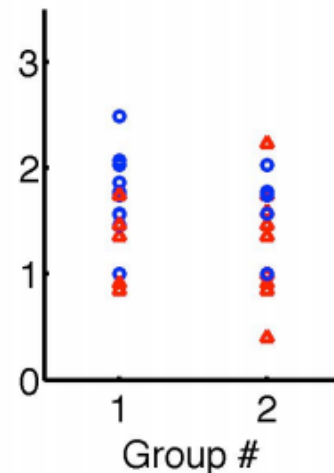
Let's start collecting them



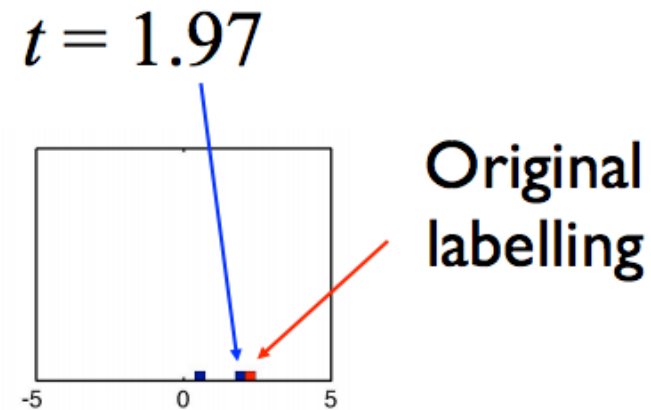
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Second re-labelling



t -value after re-labelling



And another one



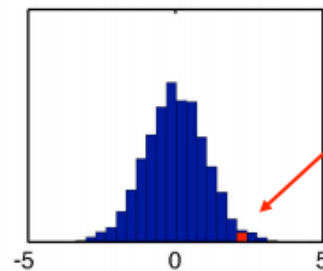
Oh dear! What now?

- We could Monte-Carlo simulate the distributions. As I did for these slides.
 - But, hinges on lots of assumptions about the data
- We could **permute** the data itself.

Of the 5000 re-labellings, only 90 had a t-value > 2.27 (the original labelling).

I.e. there is only a $\sim 1.8\%$ (90/5000) chance of obtaining a value > 2.27 if there is no difference between the groups

C.f. $p(x \geq 2.27) = 1.79\%$ for t_{18}



Original
labelling

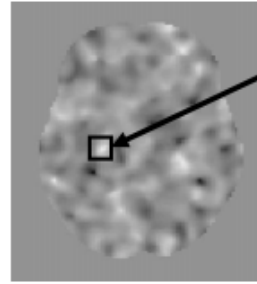
5000 re-labellings. Phew!



And now we can do the wacky stats.

We compared activation by painful stimuli in two groups of 5 subjects each.

This is what we got



Very intriguing activation. $t_8 = 4.65$

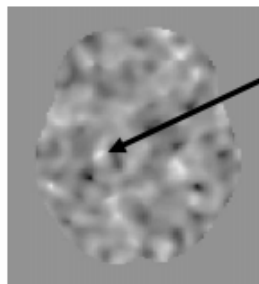
Prof. ran to write to Science. **But**, did she jump the gun?



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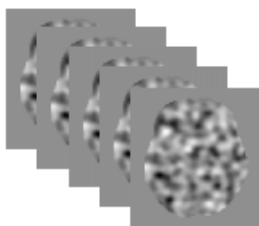
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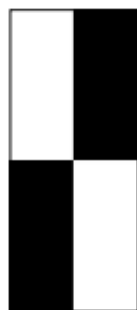
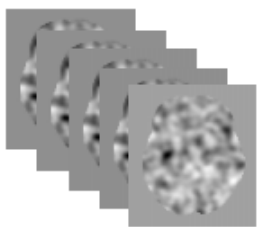
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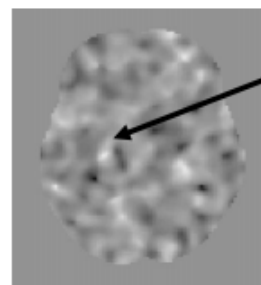
Group 1



Group 2



2nd level
model



Our group
difference map

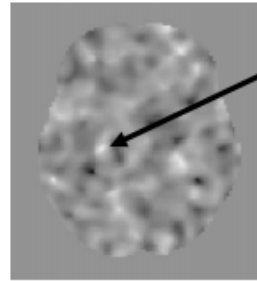
$\max(t) = 4.65$



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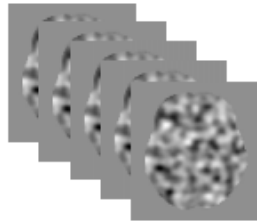
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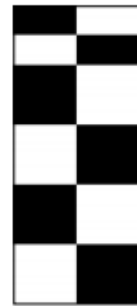
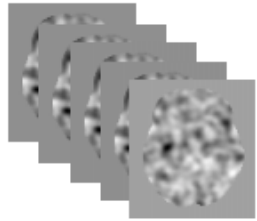
Very intriguing activation. $t_8 = 4.65$

Prof. ran to write to Science. **But**, did she jump the gun?

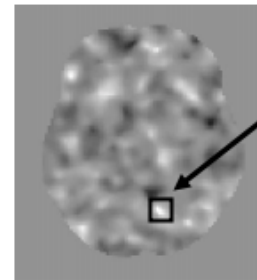
Group 1



Group 2



Permuted model



Permuted group difference map

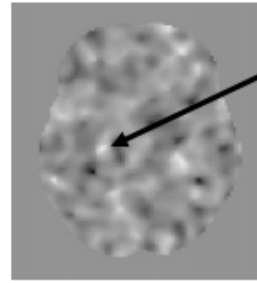
$\max(t) = 8.23$



And now we can do the wacky stats.

We compared activation by painful stimuli in two groups of 5 subjects each.

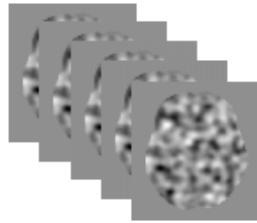
This is what we got



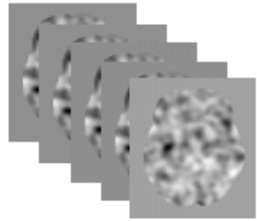
Very intriguing activation. $t_8 = 4.65$

Prof. ran to write to Science. **But**, did she jump the gun?

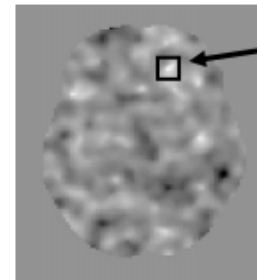
Group 1



Group 2



2nd
Permutation



2nd permuted
map

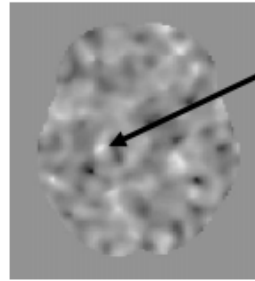
$\max(t) = 5.43$



And now we can do the wacky stats.

We compared activation by painful stimuli in two groups of 5 subjects each.

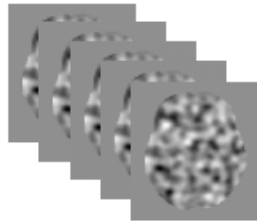
This is what we got



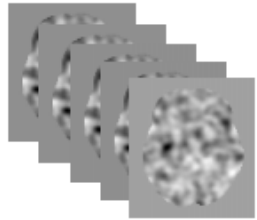
Very intriguing activation. $t_8 = 4.65$

Prof. ran to write to Science. **But**, did she jump the gun?

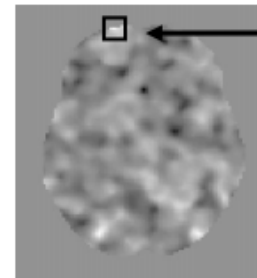
Group 1



Group 2



3rd
Permutation



3rd permuted
map

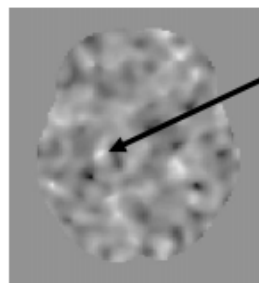
$\max(t) = 5.84$



And now we can do the wacky stats.

We compared activation by painful stimuli in two groups of 5 subjects each.

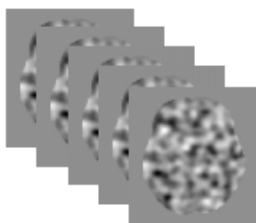
This is what we got



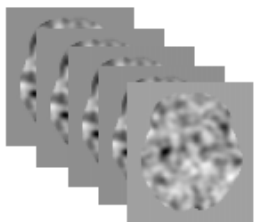
Very intriguing activation. $t_8 = 4.65$

Prof. ran to write to Science. **But**, did she jump the gun?

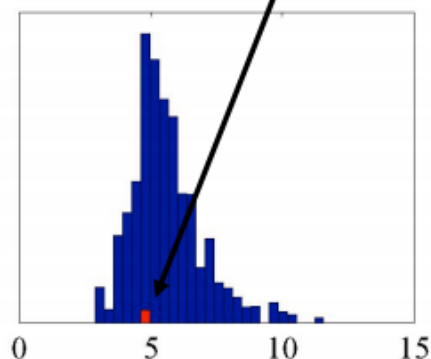
Group 1



Group 2



Original labelling



5000 permutations

3925 permutations yielded higher max(t)-value than original labelling. We **cannot** reject the null-hypothesis.

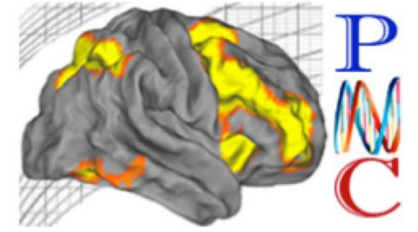


Permutations for dummies

- Needs a null-hypothesis and a test-statistic.
- + Builds its own “null-distribution” from your single data-set.
- + No assumptions about the data.
- + Can use any test-statistic, e.g. $\max(t)$, $\max(n_{\text{cluster}})$ etc.
- + Can use “classical” statistics (e.g. t -test) when data have strange distribution.
- Need to ensure exchangeability.
- Don't hold your breath.

Slides on sPBJ evaluation

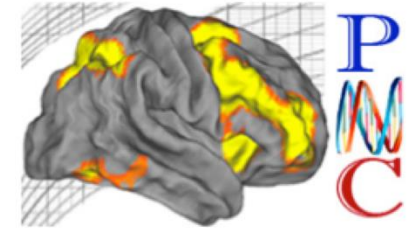
Philadelphia Neurodevelopmental Cohort (PNC)¹



- Cohort study to investigate genetic and imaging risk factors associated with psychiatric disorders
 - Led by Raquel Gur and Hakon Hakonarson
- Cross-sectional N-back data 1,600 volunteers ages 8-23

n	% female	Age (SD)	Age range
1000	55%	14.7 (3.5)	8-23

¹Satterthwaite et al., *Neuroimage*, 2016



N-back data analysis

- We analyzed a subset of 1000 subjects imaged as part of the Philadelphia Neurodevelopmental Cohort
- **Our scientific goal is to understand how activation associated with increasing working memory load is related to task performance**

$$Y_i(v) = \alpha_0(v) + \alpha_1(v) \times \text{sex}_i + \alpha_2(v) \times \text{age}_i + \alpha_3(v) \times \text{mot}_i \\ + \beta(v) \times d'_i + \epsilon_i(v)$$

- Our null hypothesis at v is $H_0(v): \beta(v) = 0$

Satterthwaite et al., *J Neuro*, 2013

Simulating realistic heteroskedasticity

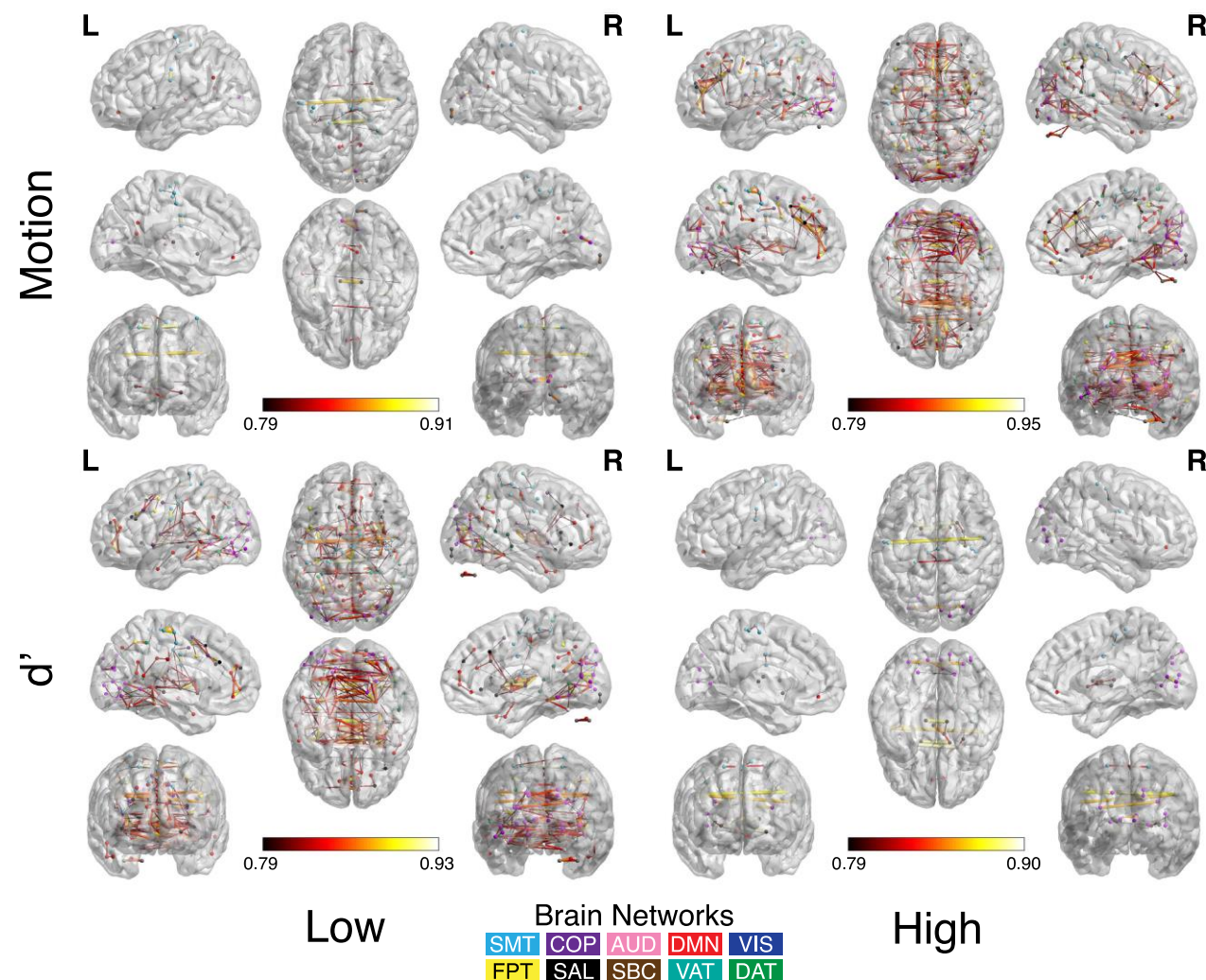
- To generate realistic heteroskedasticity we bootstrapped from N-back data
- In a subset of 1000 subject from the PNC, we fit the model

$$Y_i(v) = \alpha_0(v) + \alpha_1(v) \times \text{sex}_i \\ + f(v, \text{age}_i) + g(v, \text{mot}_i) + h(v, d'_i) + \epsilon_i(v)$$

- The sample mean of $\hat{\epsilon}_i(v)$ is then independent of the covariates
- The covariance function $\text{Cov}\{\hat{\epsilon}_i(v), \hat{\epsilon}_i(w)\}$ may be affected by covariates

The covariance function is affected by covariates in the N-back sample

- $\text{Cov}\{Y_i(v), Y_i(w) | X_i\} \neq \text{Cov}\{Y_j(v), Y_j(w) | X_j\}$ implies heteroskedasticity



Intuition behind simulation procedure

- Treat residuals $\hat{\epsilon}_i(v)$ of 1000 subjects as population under the null
- In each simulation draw a bootstrap sample and fit the model

$$\hat{\epsilon}_{ib}(v) = \alpha_{0b}(v) + \alpha_{1b}(v) \times \text{sex}_{ib} + \alpha_{2b}(v) \times \text{age}_{ib} + \alpha_{3b}(v) \times \text{mot}_{ib} \\ + \beta_b(v) \times d'_{ib} + \epsilon_{ib}(v)$$

- Perform test of $H_0(v): \beta_b(v) = 0$ and $H_0(v): \alpha_{3b}(v) = 0$ to assess effect of heteroskedasticity on FWER of SEI procedures

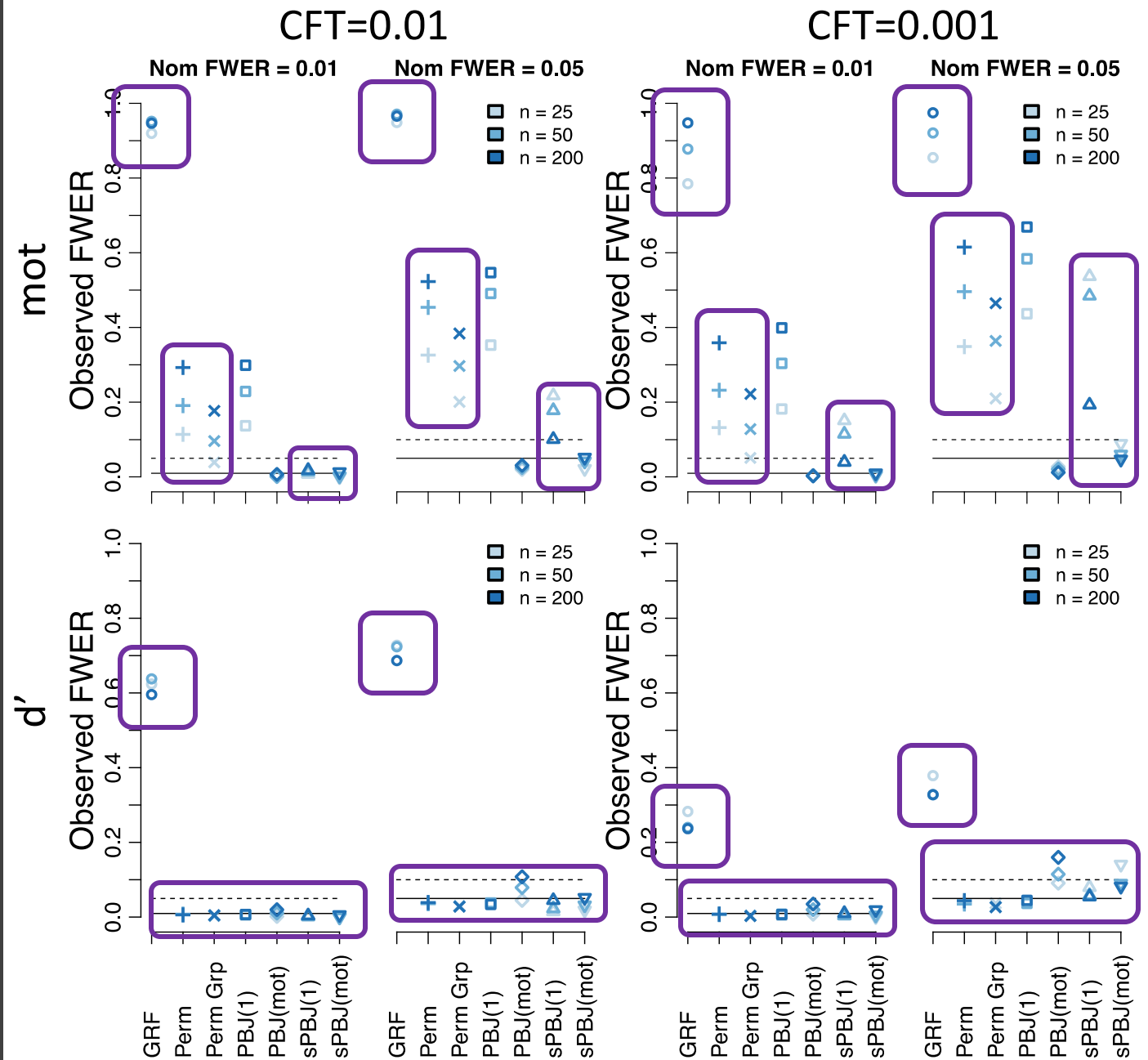
Procedures we will compare

- GRF - classical GRF based method assuming local correlation in field
- Perm - permutation procedure assumes exchangeability
- Perm Grp - permutation procedure, attempts to adjust for non exchangeability
- PBJ - parametric bootstrap joint procedure [PBJ(1), PBJ(mot)] 
- sPBJ - semiparametric bootstrap joint procedure [sPBJ(1), sPBJ(mot)]

Friston et al., *Neuroimage*, 1996
Winkler et al., *Neuroimage*, 2014
Winkler et al., *Neuroimage*, 2015
Vandekar et al., *Biometrics*, 2019

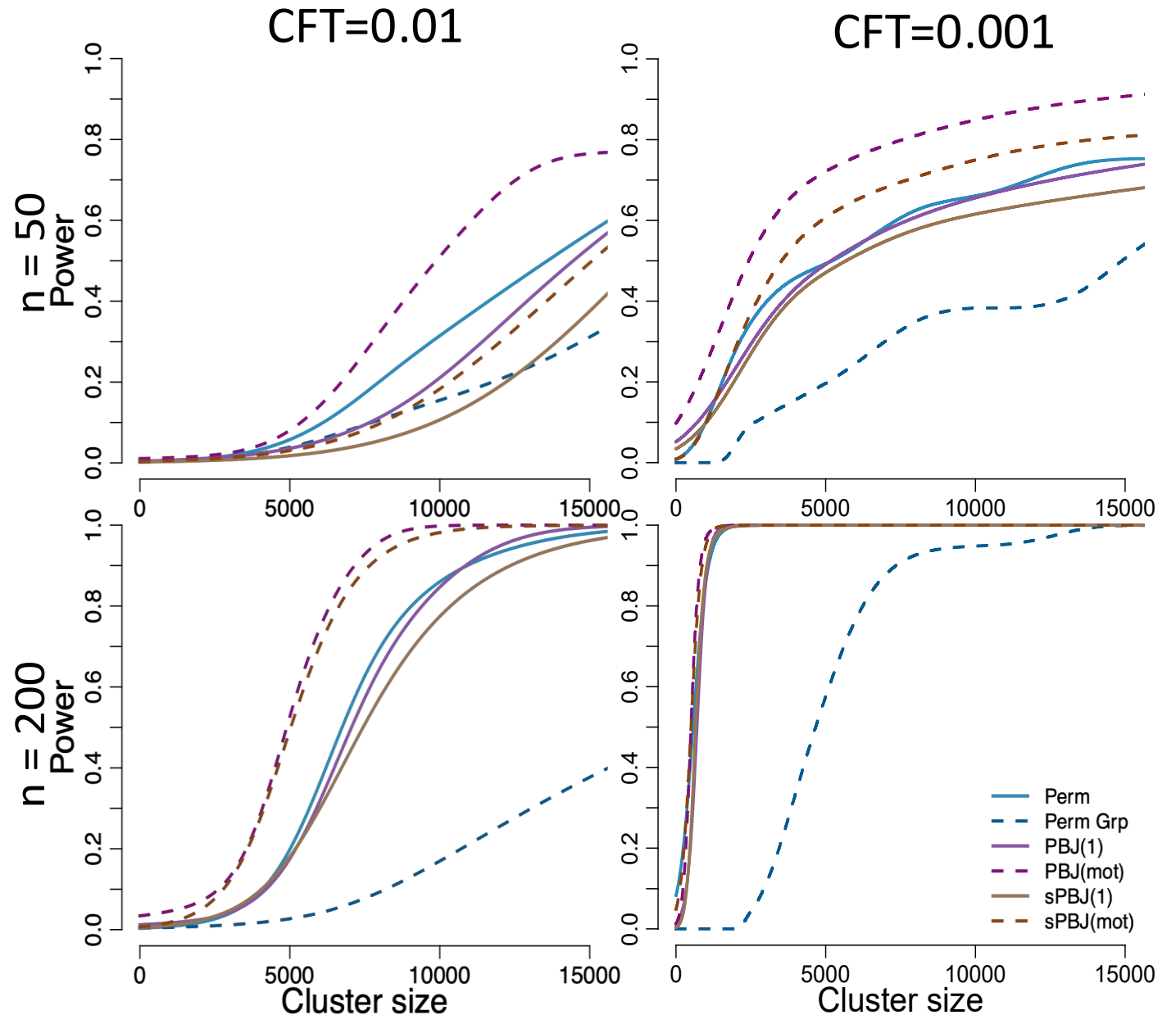
Heteroskedasticity simulation: type 1 error rates

- For the test of the motion covariate
 - GRF near 100% error rate
 - Permutations and unweighted PBJ have inflated error rate
 - Robust methods have near nominal performance at less conservative CFTs
- For the test of the d' covariate
 - GRF still quite high
 - Other methods near or approach nominal level
- Next slide: power analysis



Heteroskedasticity simulation: power results

- Power results for the test of the d' covariate for a rejection threshold of 0.01
 - Type 1 error rates near the nominal level
- Motion deweighting may improve power here

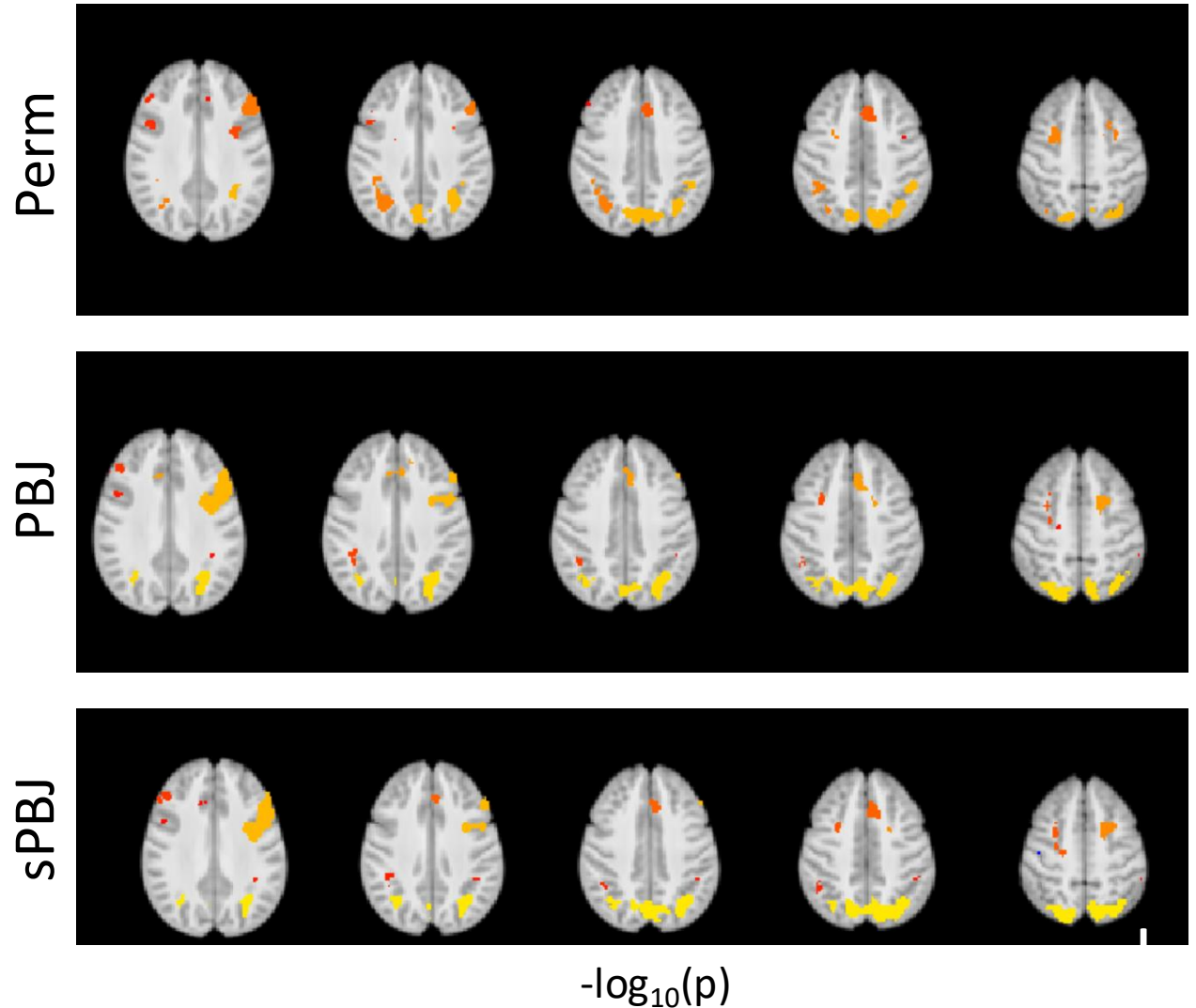


N-back data analysis

- We used a random subset of 200 subjects to evaluate the different procedures
- Recall our goal: **to identify regions where WM performance is associated with WM activation**
- The PBJ and sPBJ procedures used voxel-wise weights proportional to the inverse of the subject variance image estimates.
- Use SEI to compute p-values for each cluster

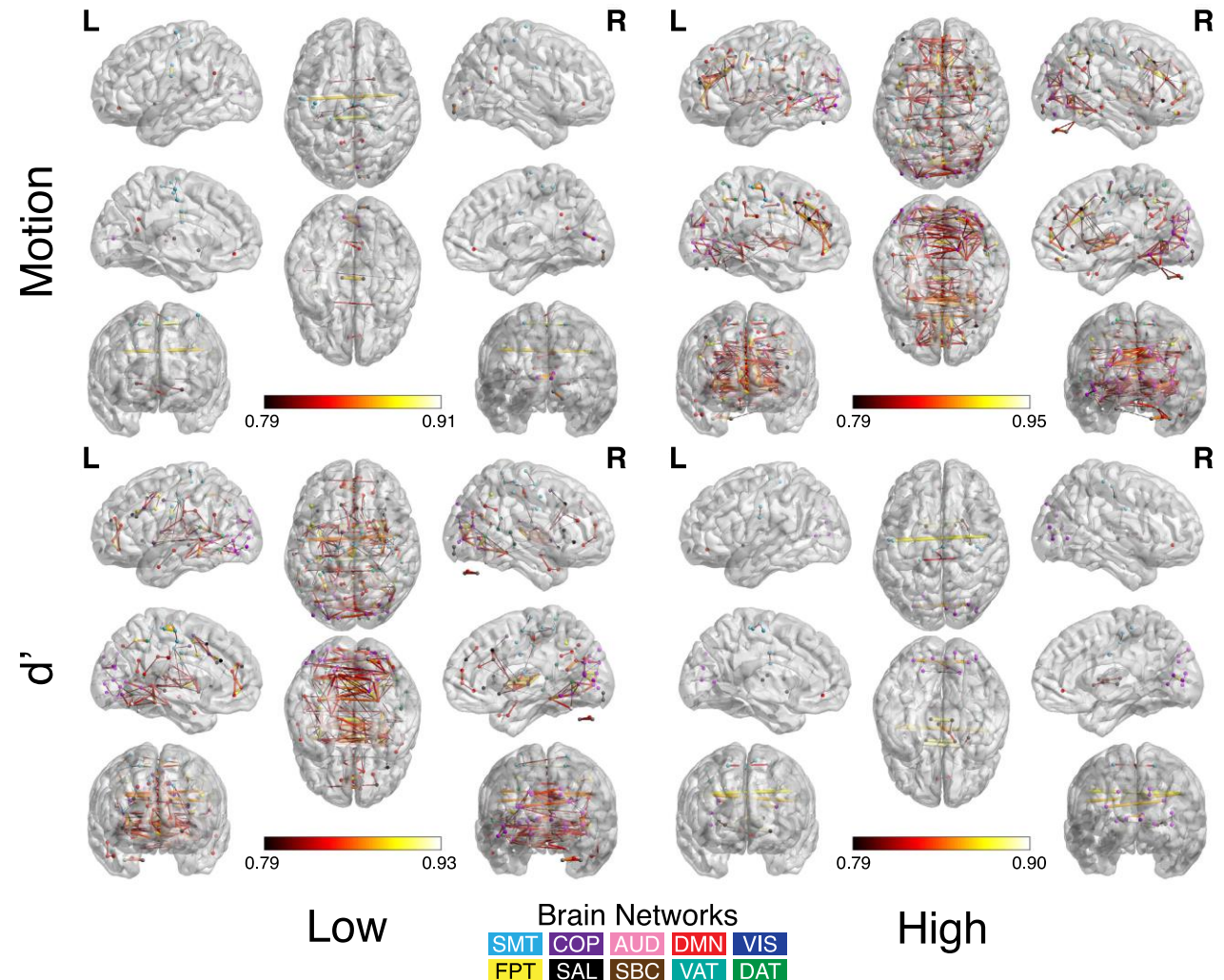
N-back data analysis results

- The sPBJ have smaller p-values in some regions
- Small p-values indicate cluster sizes that are unlikely under the global null $H_0(v): \beta_b(v) = 0$, for all v



The covariance function is affected by covariates in the N-back sample

- $\text{Cov}\{Y_i(v), Y_i(w) | X_i\} \neq \text{Cov}\{Y_j(v), Y_j(w) | X_j\}$ implies heteroskedasticity
- Functional connectivity is evidence of heteroskedasticity



Comparing assumptions



	GRF	FLAME	Permutation	sPBJ
Assumes homoskedasticity	😞	😄 / 😞	😄 / 😞	😄
Uses normal approximations	😞	😄	😄	😞 / 😄
Repeated measurements	😄	😄	😞 / 😄	😄
Robust to model misspecification	😞	😞	😞	😄
Uses some kind of approximation	😞	😄 / 😞	😞	😞

*Methods overlap software packages, e.g. permutation is available in SPM and AFNI, but not as the default

TFCE

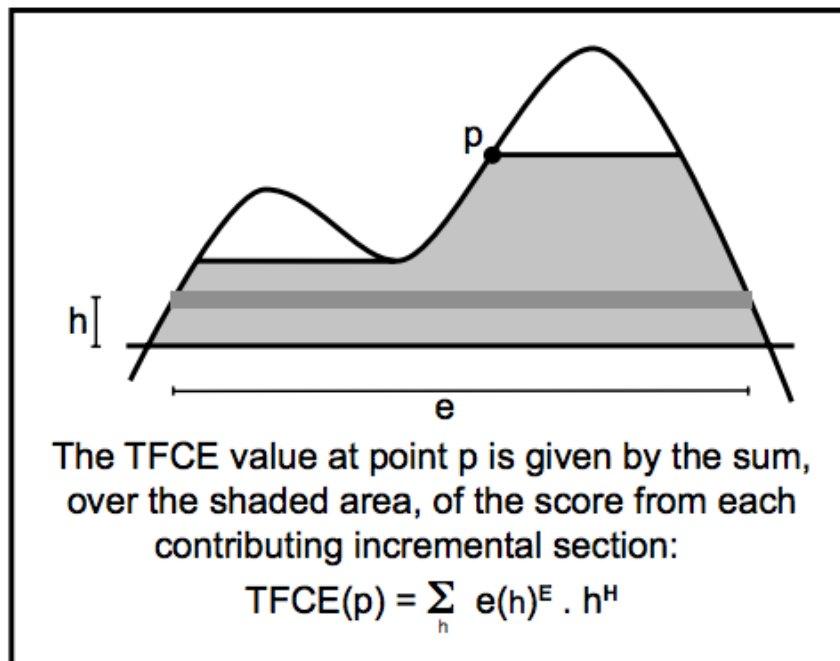


TFCE

Threshold-Free Cluster Enhancement

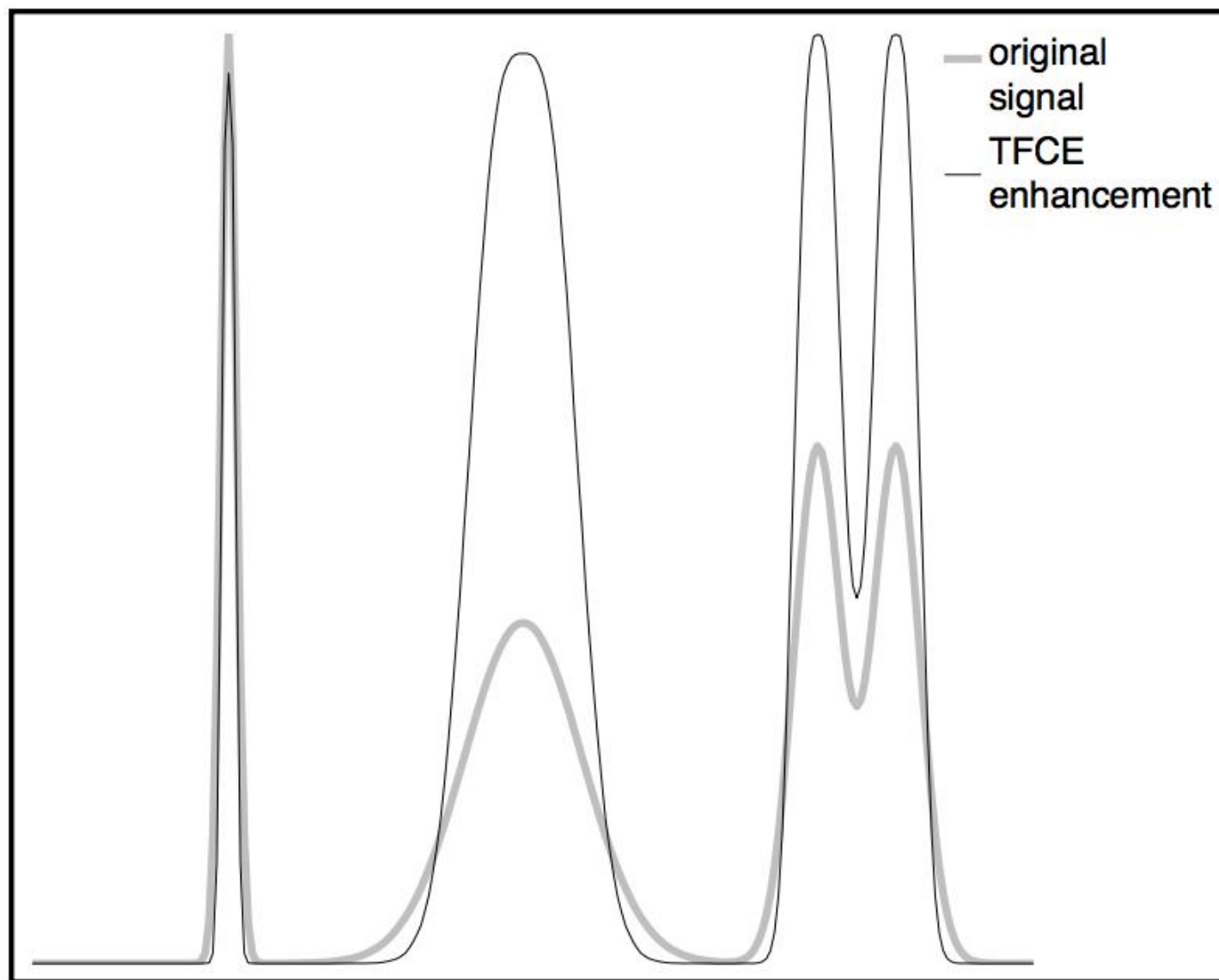
[Smith & Nichols, NeuroImage 2009]

- Cluster thresholding:
 - popular because it's sensitive, due to its use of spatial extent
 - but the pre-smoothing extent is arbitrary
 - and so is the cluster-forming threshold
 - ➔ unstable and arbitrary
- TFCE
 - integrates cluster “scores” over all possible thresholds
 - output at each voxel is measure of local cluster-like support
 - similar sensitivity to optimal cluster-thresholding, but stable and non-arbitrary





Qualitative example





TFCE for FSL-VBM

